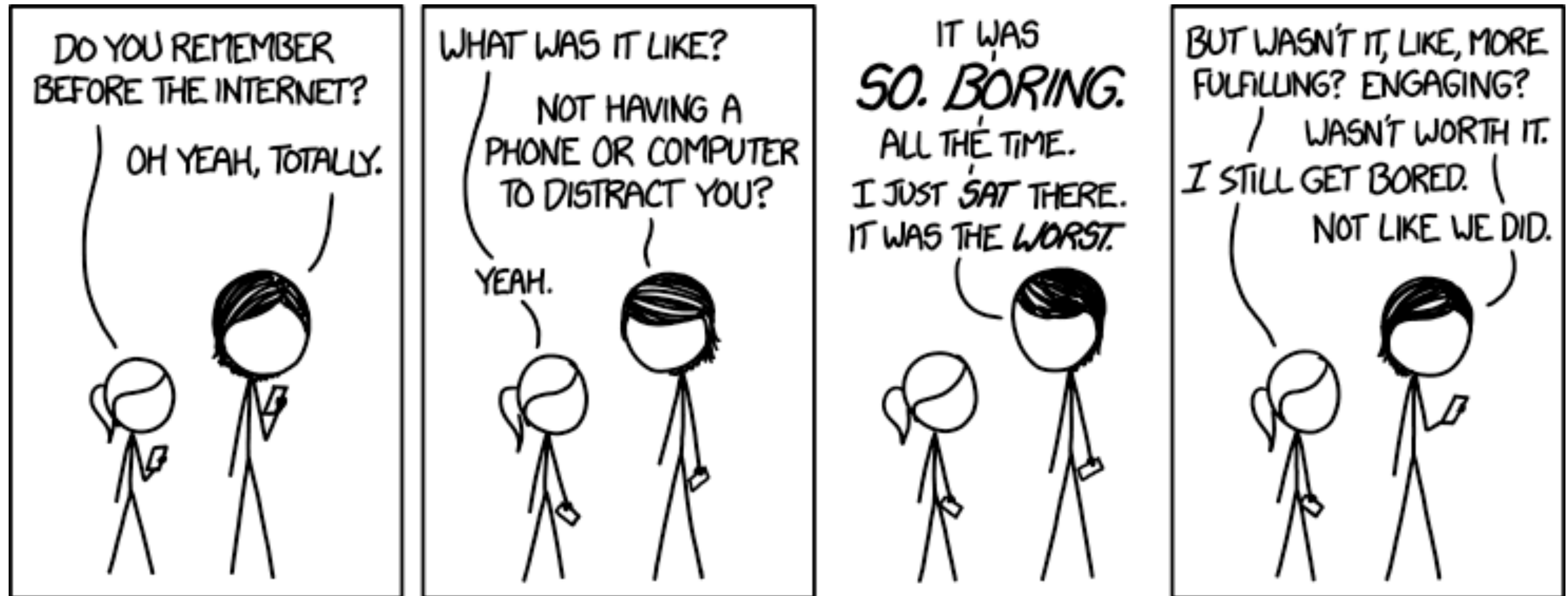


The Network Layer



Learning Goals

- ▶ Get an understanding of routing on the network layer
 - ▶ Input port functions
 - ▶ Longest prefix matching
 - ▶ Buffer management
 - ▶ Scheduling policies
- ▶ Understand the Internet Protocol and network addressing
 - ▶ IPv4 basics
 - ▶ Addressing & subnets
 - ▶ DHCP
 - ▶ NAT
 - ▶ IPv6

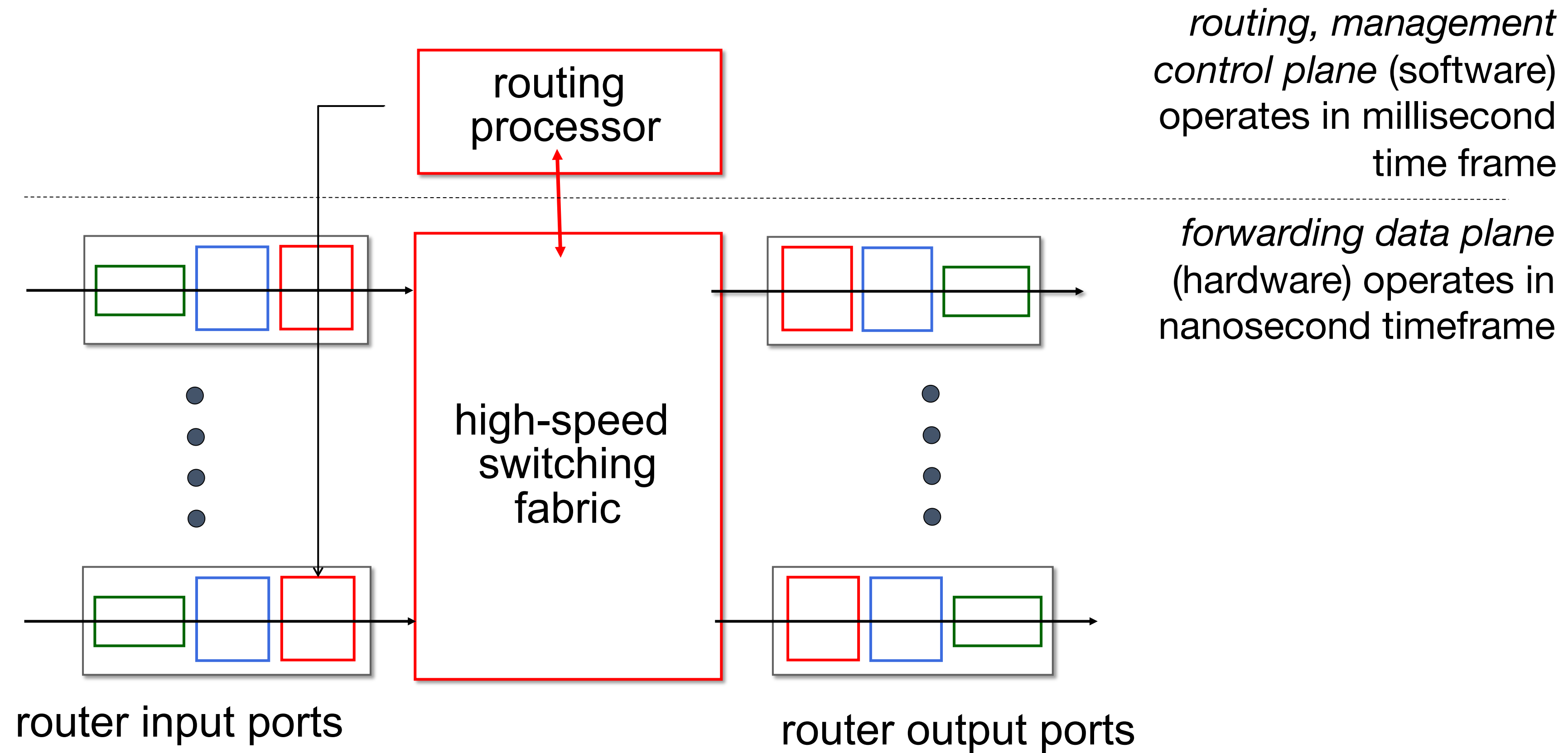
Learning Goals

- ▶ Understand how IPsec can be used to establish a VPN
 - ▶ IPsec basics
 - ▶ Security Associations
 - ▶ Security Policy & Security Association Databases
 - ▶ Internet Key Exchange

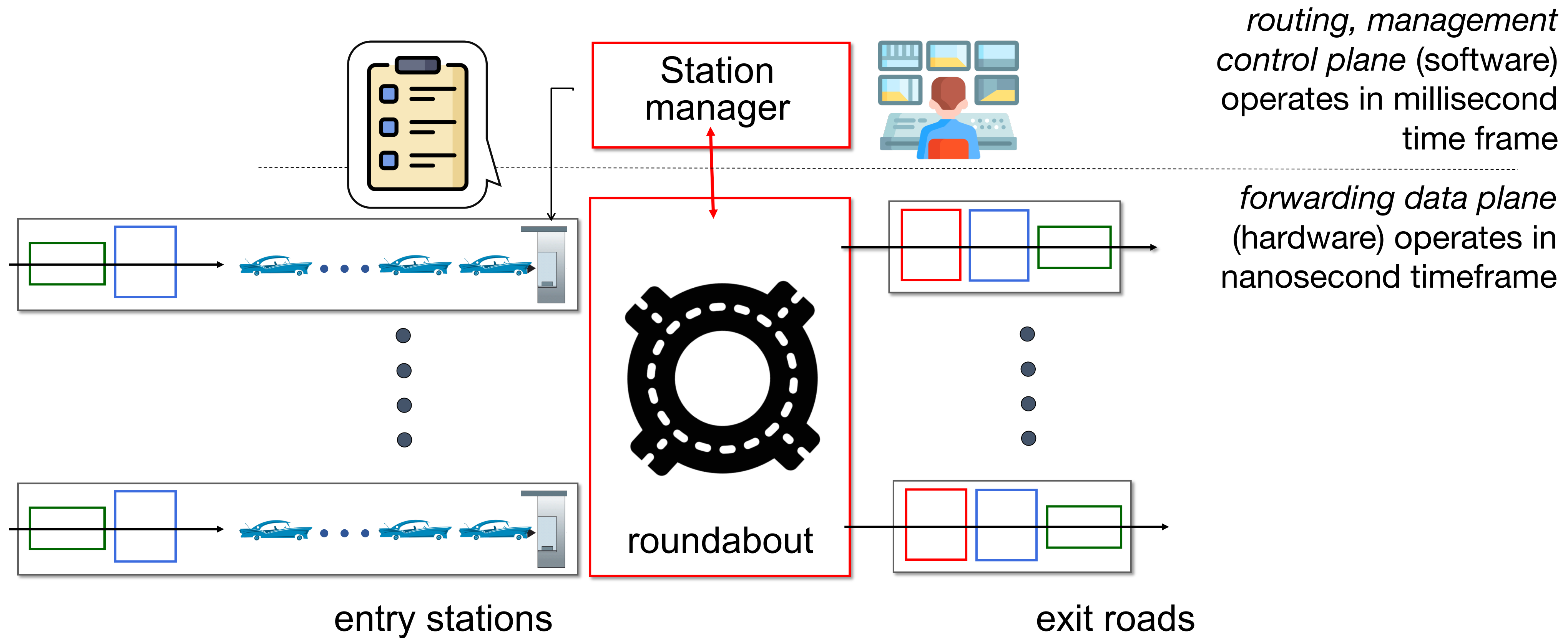
Learning Goal 50AEF3

How does network layer routing work?

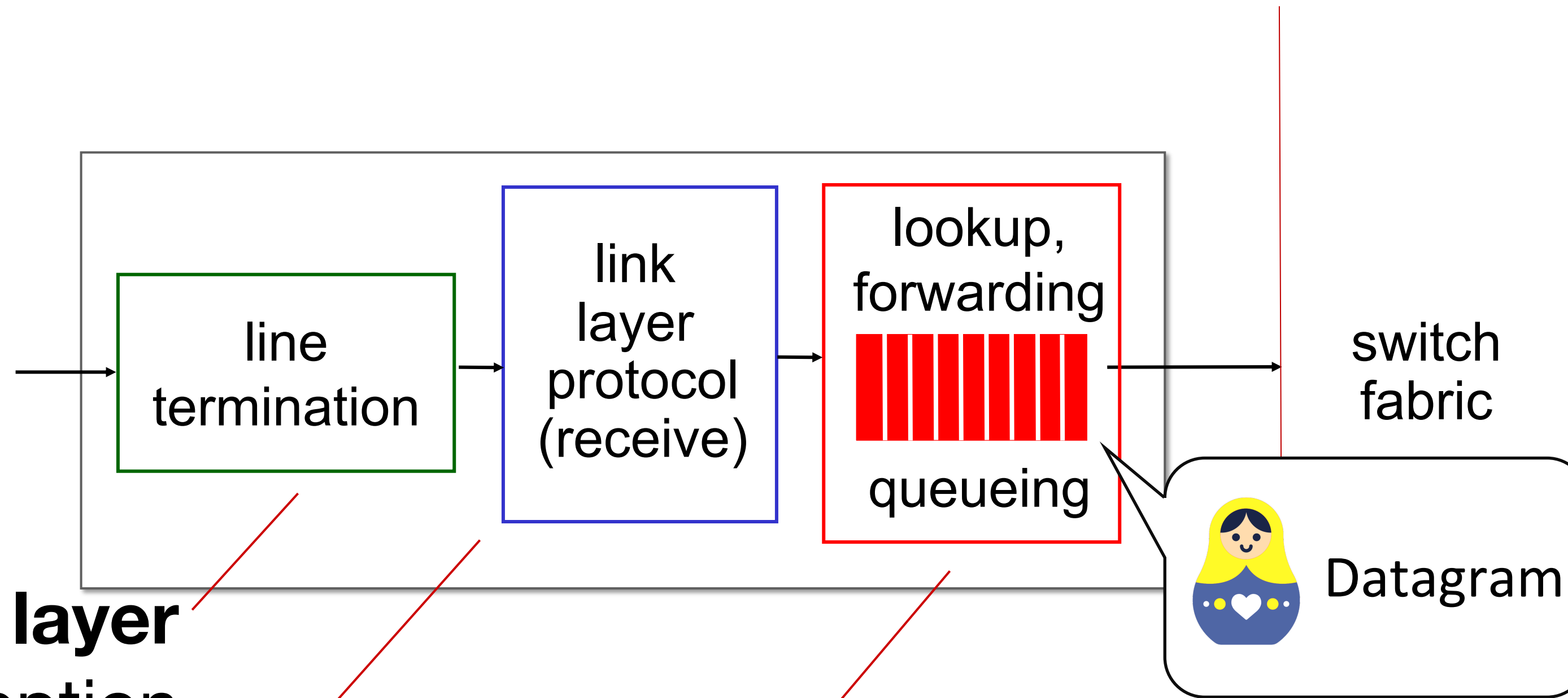
High-level overview of generic router architecture



Input port functions – Back to the cars analogy

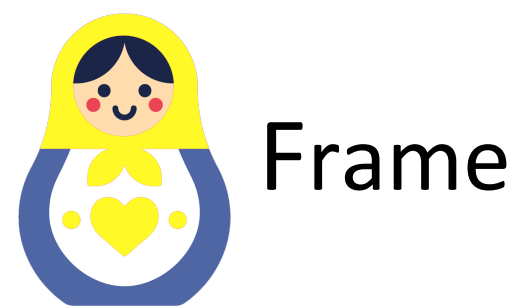


Input port functions



Physical layer
Bit-level reception

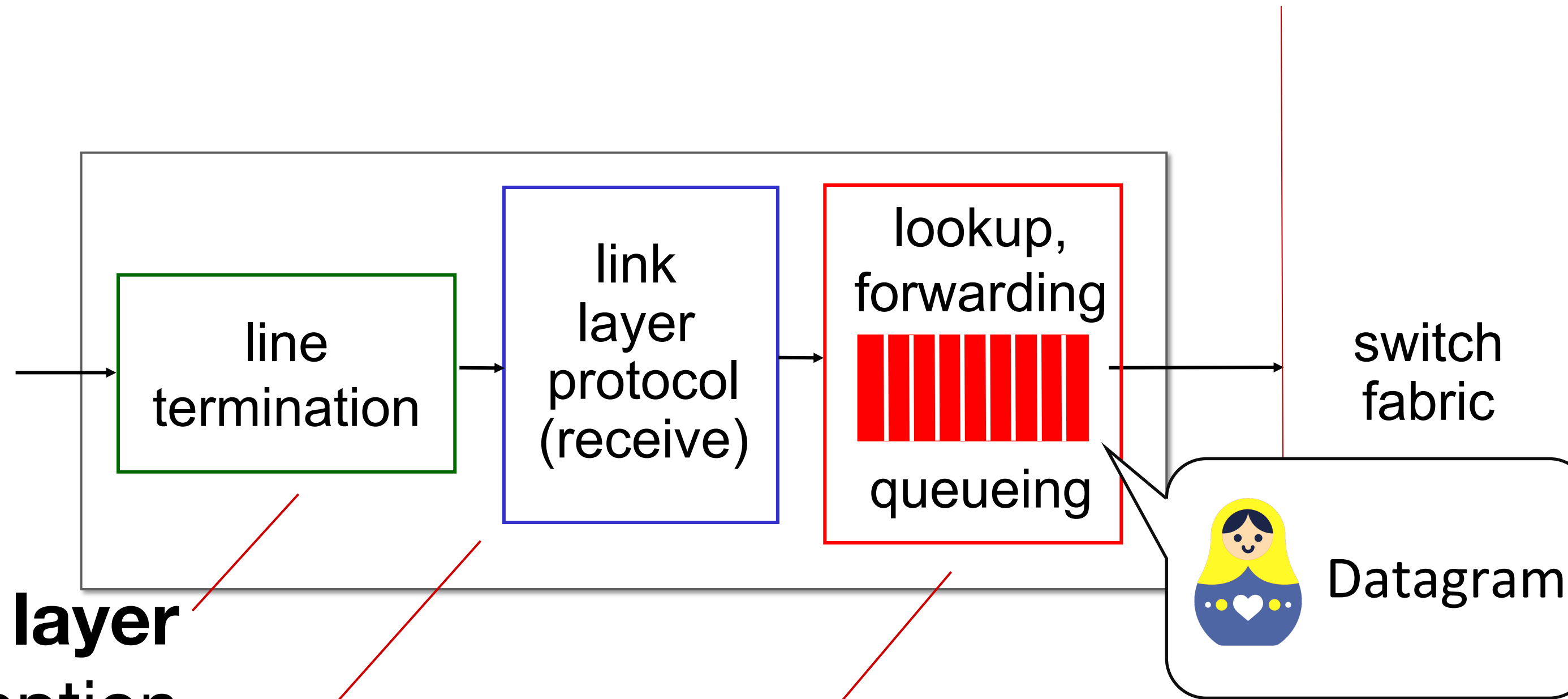
Link layer
e.g., Ethernet



Decentralized Switching

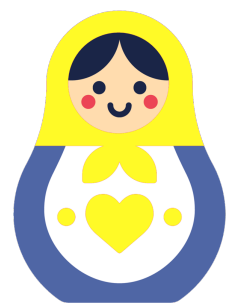
- ▶ **Goal:** complete input port processing at 'line speed'
- ▶ Using header field values, lookup output port using forwarding table in input port memory ("match plus action")
- ▶ Input port queuing: if datagrams arrive faster than forwarding rate into switch fabric

Input port functions



Physical layer
Bit-level reception

Link layer
e.g., Ethernet



Frame

Decentralized Switching

- ▶ Using header field values, lookup output port using forwarding table in input port memory (“match plus action”)
- ▶ Destination-based forwarding: forward based only on destination IP address (traditional)
- ▶ Generalized forwarding: forward based on any set of header field values

Destination-based forwarding

<i>forwarding table</i>	
Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010000 00000100	n 3
11001000 00010111 00010000 00000111	
11001000 00010111 00011000 11111111	
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

Q: but what happens if ranges don't divide up so nicely?

Longest prefix matching

Longest prefix match

When looking for forwarding table entry for given destination address, use *longest* address prefix that matches destination address.

Destination Address Range	Link interface
11001000 00010111 00010*** *****	0
11001000 00010111 00011000 *****	1
11001000 00010111 00011*** *****	2
otherwise	3

examples:

11001000 00010111 00010110 10100001 which interface?

11001000 00010111 00011000 10101010 which interface?

Longest prefix matching

Longest prefix match

When looking for forwarding table entry for given destination address, use *longest* address prefix that matches destination address.

Destination Address Range	Link interface
11001000 00010111 00010*** *****	0
11001000 00010111 00011000 *****	1
11001000 match! 00011*** *****	2
otherwise	3

examples:

11001000 00010111 00010110 10100001 which interface?

11001000 00010111 00011000 10101010 which interface?

Longest prefix matching

Longest prefix match

When looking for forwarding table entry for given destination address, use *longest* address prefix that matches destination address.

Destination Address Range	Link interface
11001000 00010111 00010*** *****	0
11001000 00010111 00011000 *****	1
11001000 00010111 00011*** *****	2
otherwise	3

match!

examples:

11001000 00010111 00010110 10100001	which interface?
11001000 00010111 00011000 10101010	which interface?

Longest prefix matching

Longest prefix match

When looking for forwarding table entry for given destination address, use *longest* address prefix that matches destination address.

Destination Address Range	Link interface
11001000 00010111 00010*** *****	0
11001000 00010111 00011000 *****	1
11001000 00010111 00011*** *****	2
otherwise	3

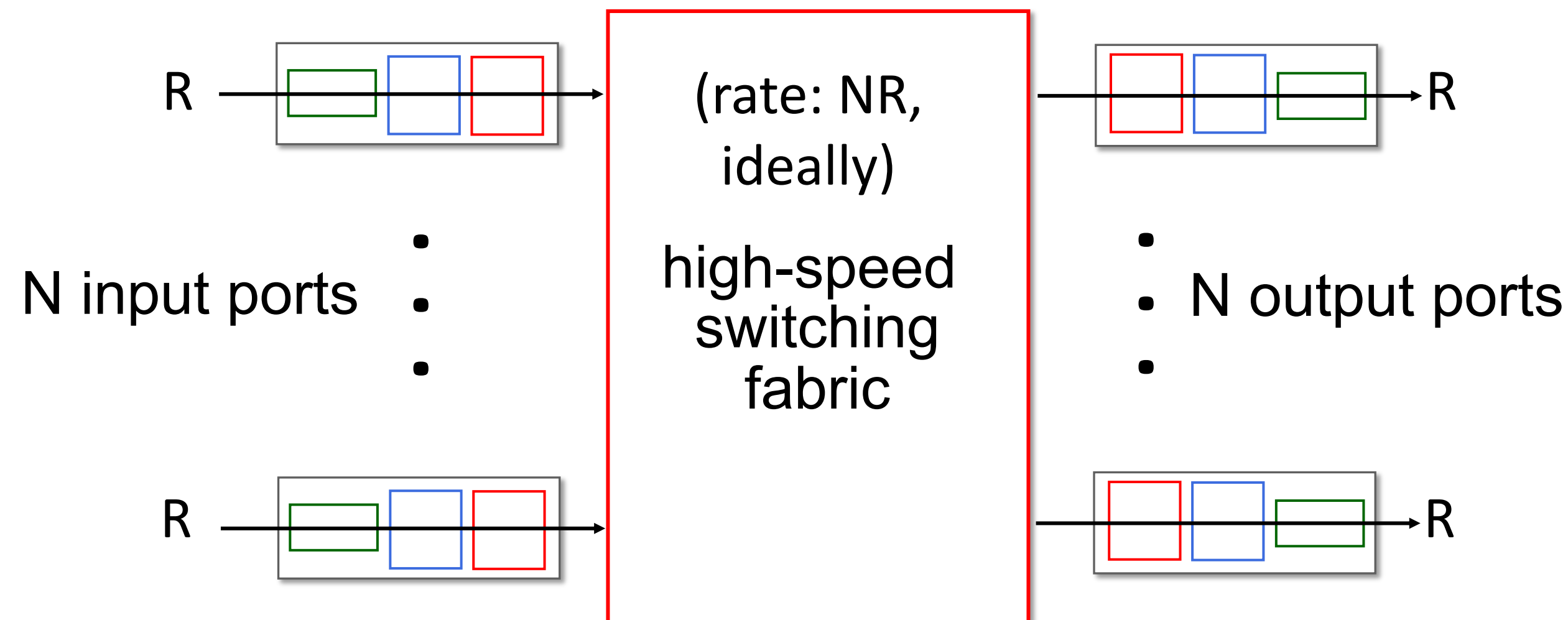
match!

examples:

11001000 00010111 00010110 10100001	which interface?
11001000 00010111 00011000 10101010	which interface?

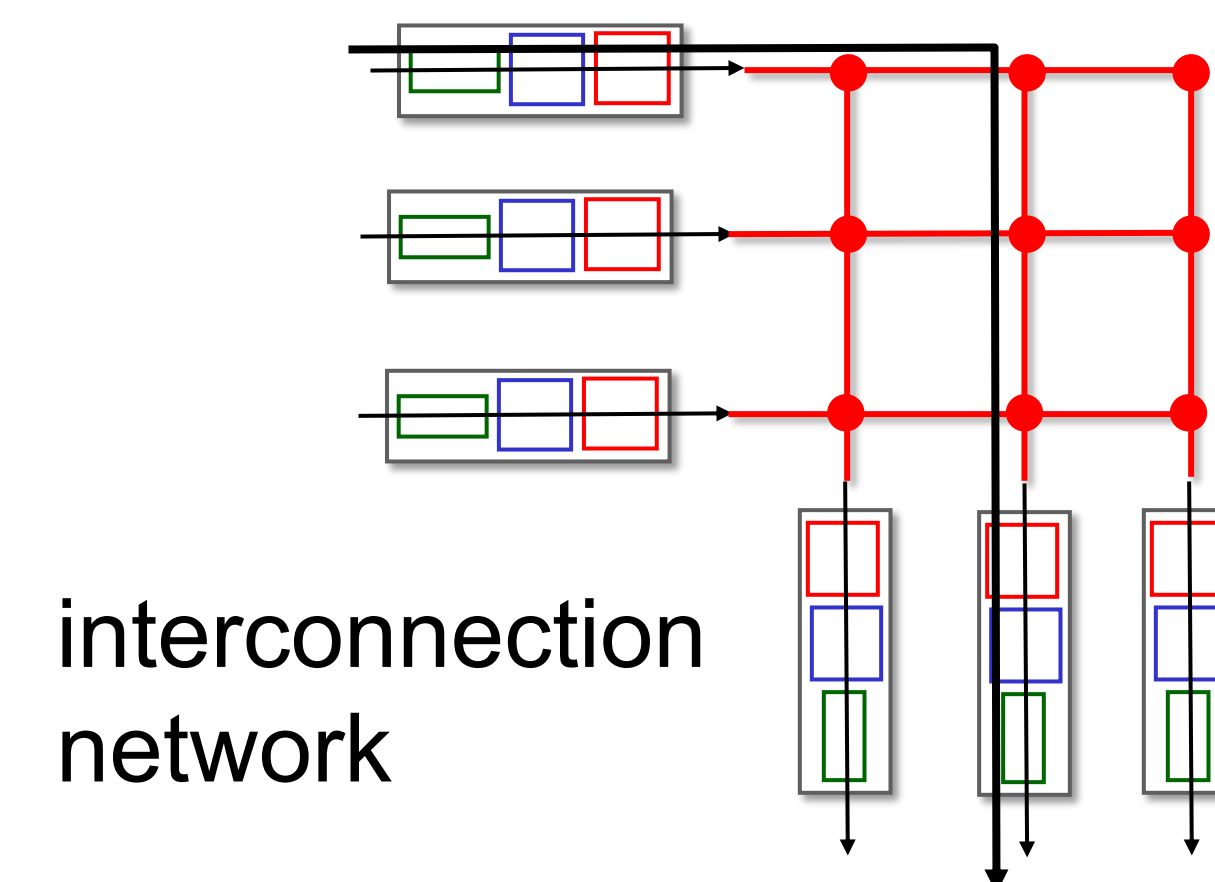
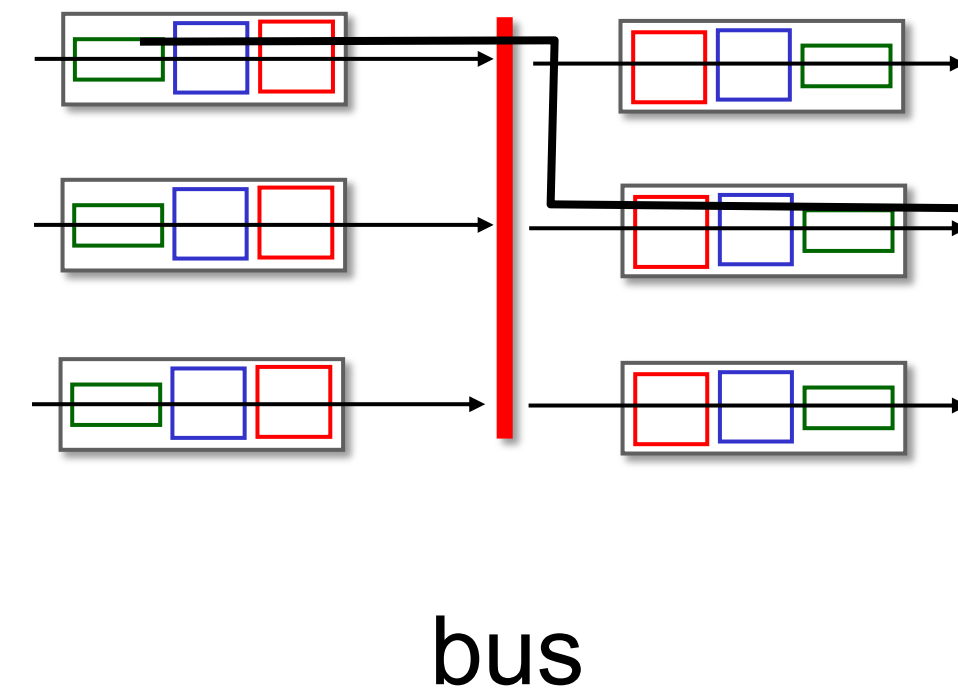
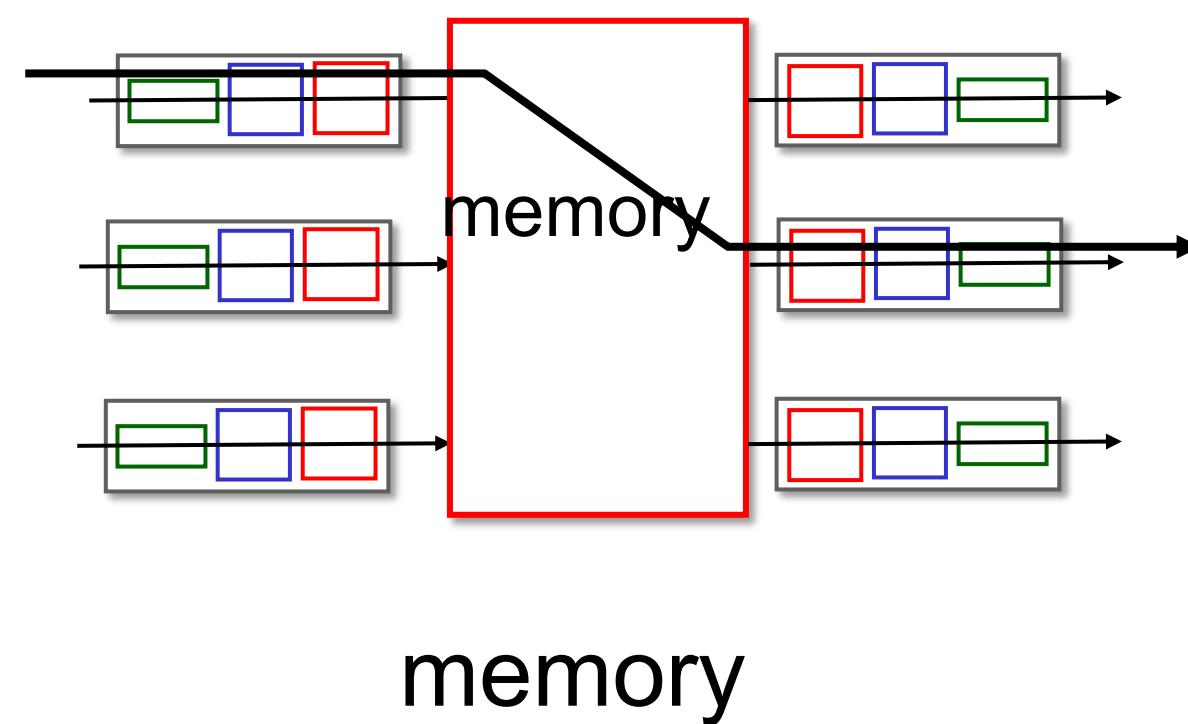
Switching fabrics

- ▶ Transfer packet from input link to appropriate output link
- ▶ **Switching rate:** rate at which packets can be transferred from inputs to outputs
- ▶ Often measured as multiple of input/output line rate
- ▶ N inputs: switching rate N times line rate desirable



Switching fabrics

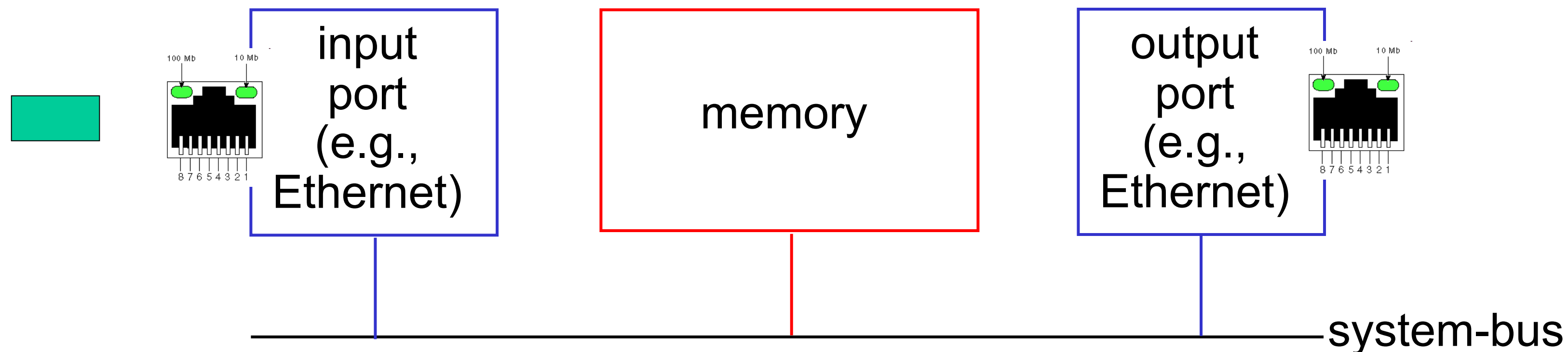
- ▶ Transfer packet from input link to appropriate output link
- ▶ **Switching rate:** rate at which packets can be transferred from inputs to outputs
 - ▶ Often measured as multiple of input/output line rate
 - ▶ N inputs: switching rate N times line rate desirable
- ▶ Three major types of switching fabrics



Adapted from slides by J.F Kurose and K.W. Ross

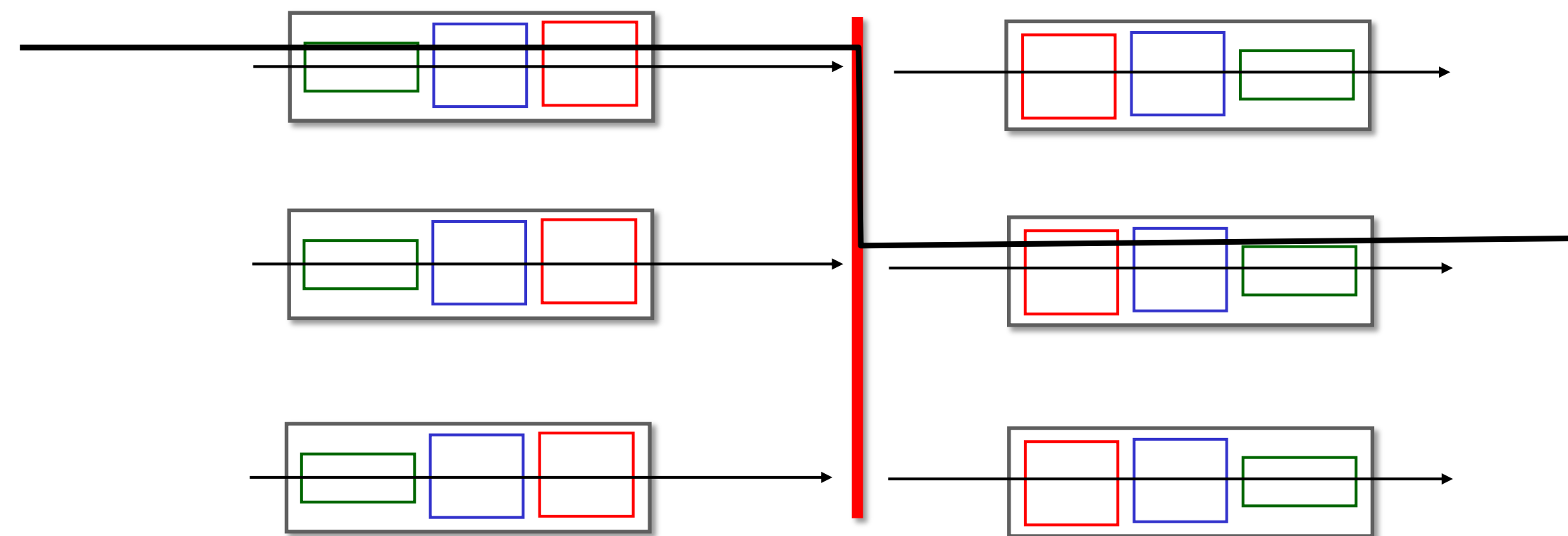
Switching via memory

- ▶ First generation routers:
 - ▶ Traditional computers with switching under direct control of CPU
 - ▶ Packet copied to system's memory
 - ▶ Speed limited by memory bandwidth (2 system-bus crossings per datagram)



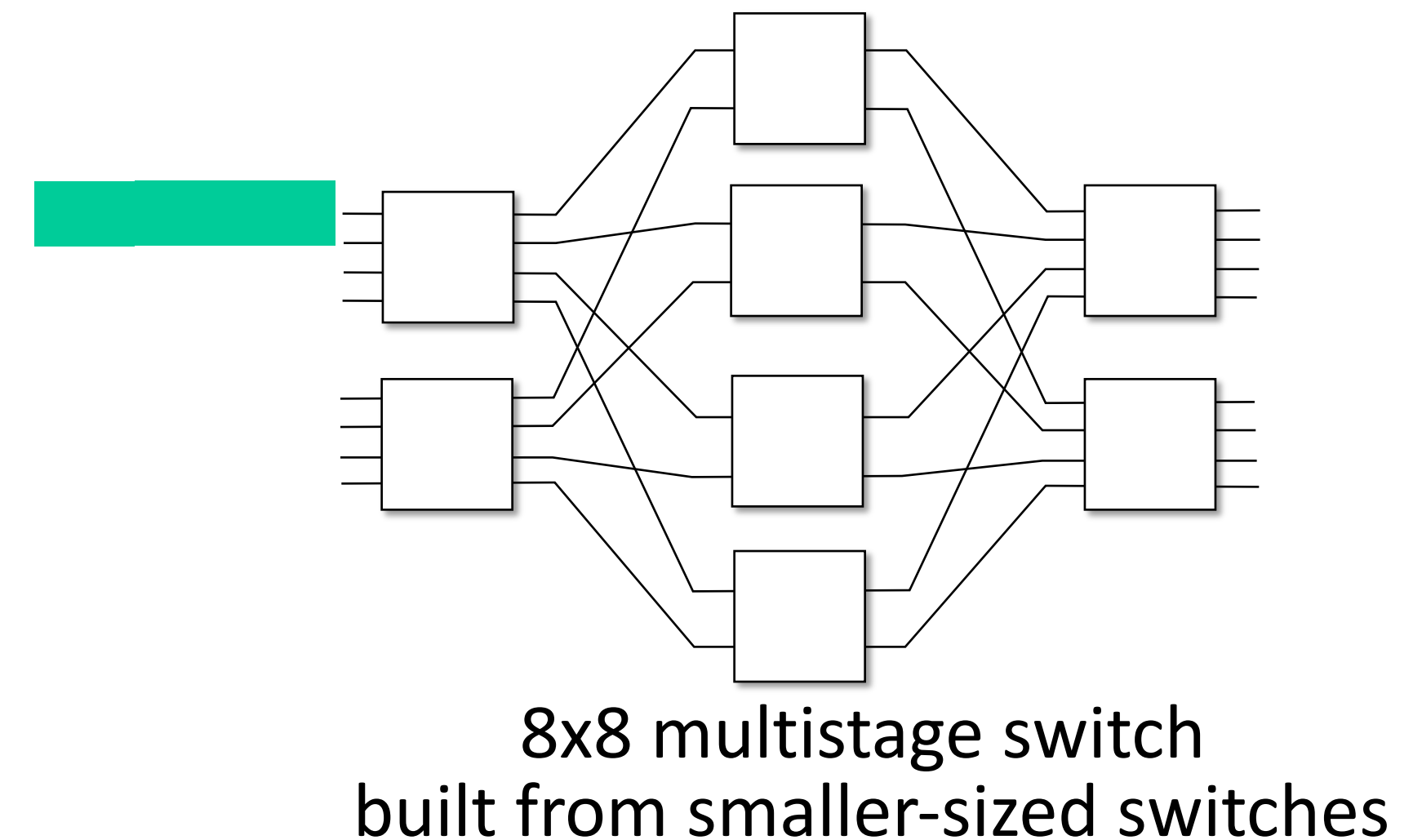
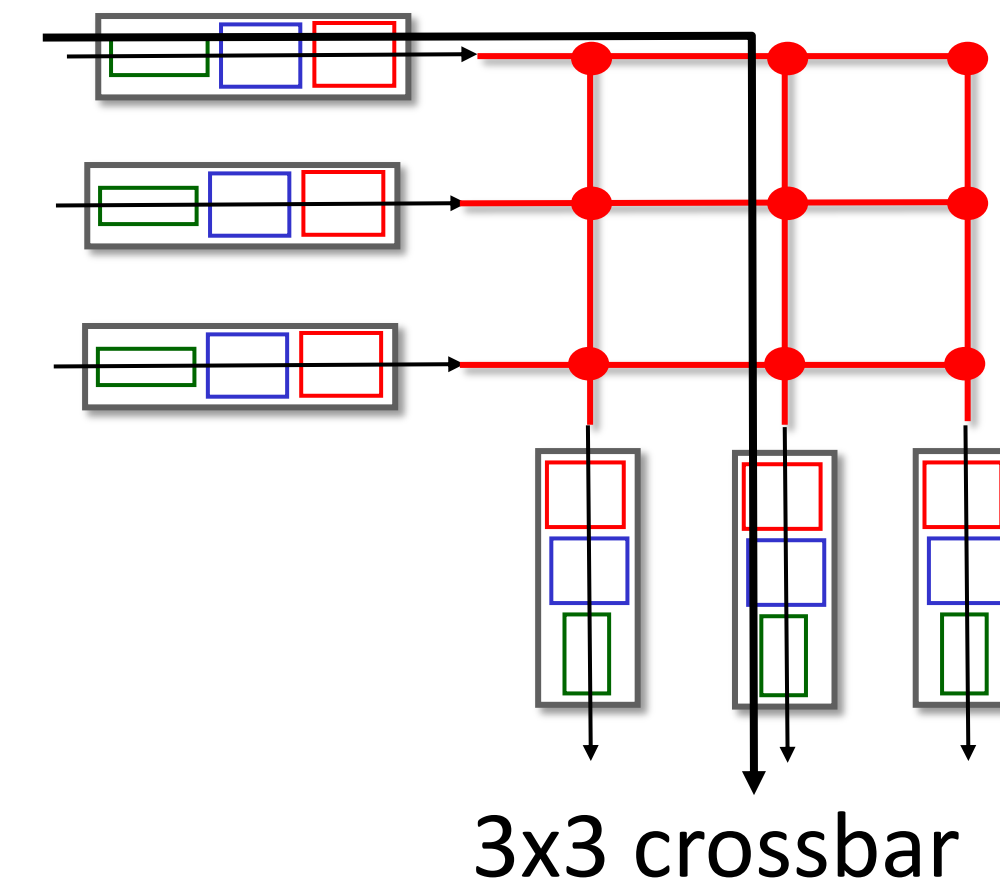
Switching via a bus

- ▶ Datagram from input port memory to output port memory via a shared bus
- ▶ Bus contention: switching speed limited by bus bandwidth
- ▶ Incoming packets are broadcasted with additional headers which are removed before sending it on



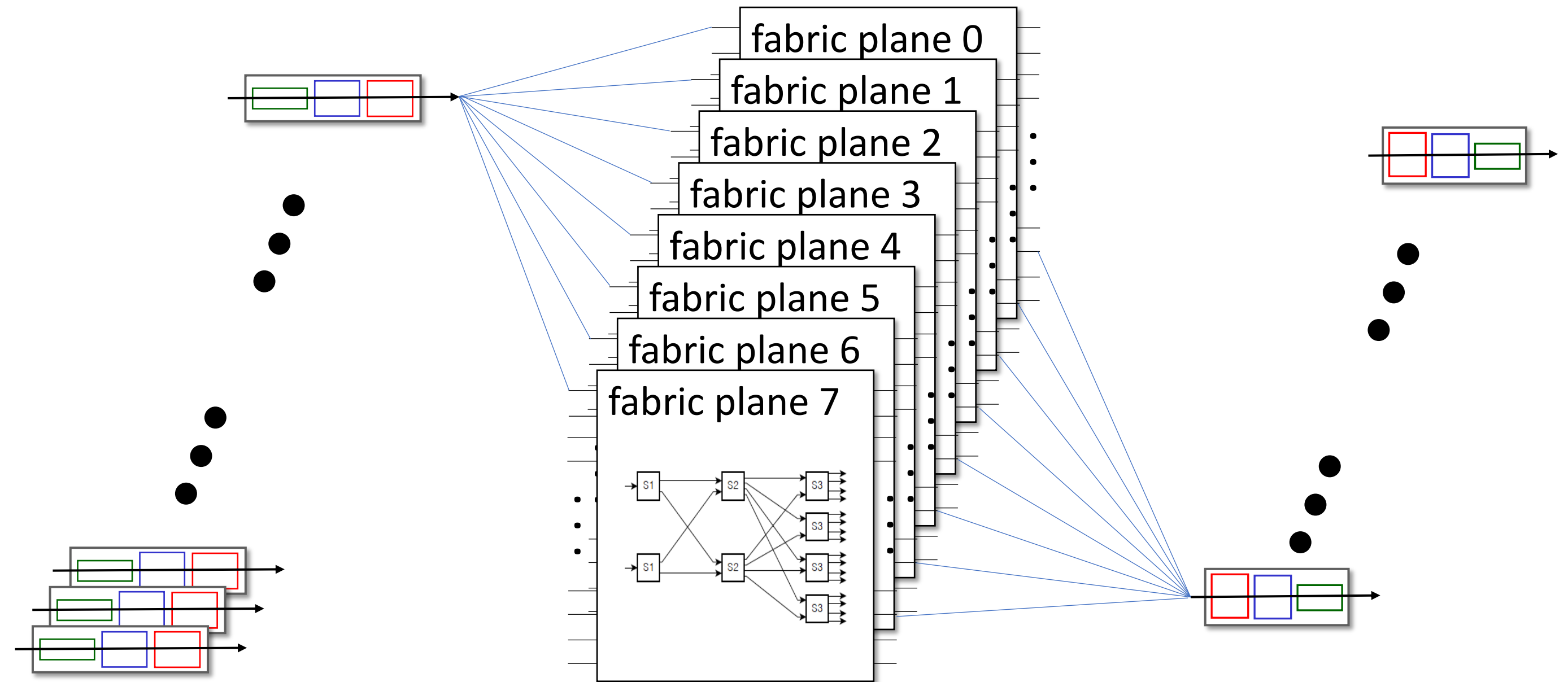
Switching via interconnection network

- ▶ Crossbar networks initially developed to connect processors in multiprocessor
- ▶ Multistage switch: $n \times n$ switch from multiple stages of smaller switches
- ▶ Exploiting parallelism
 - ▶ Fragment datagram into fixed length cells on entry
 - ▶ Switch cells through the fabric, reassemble datagram at exit



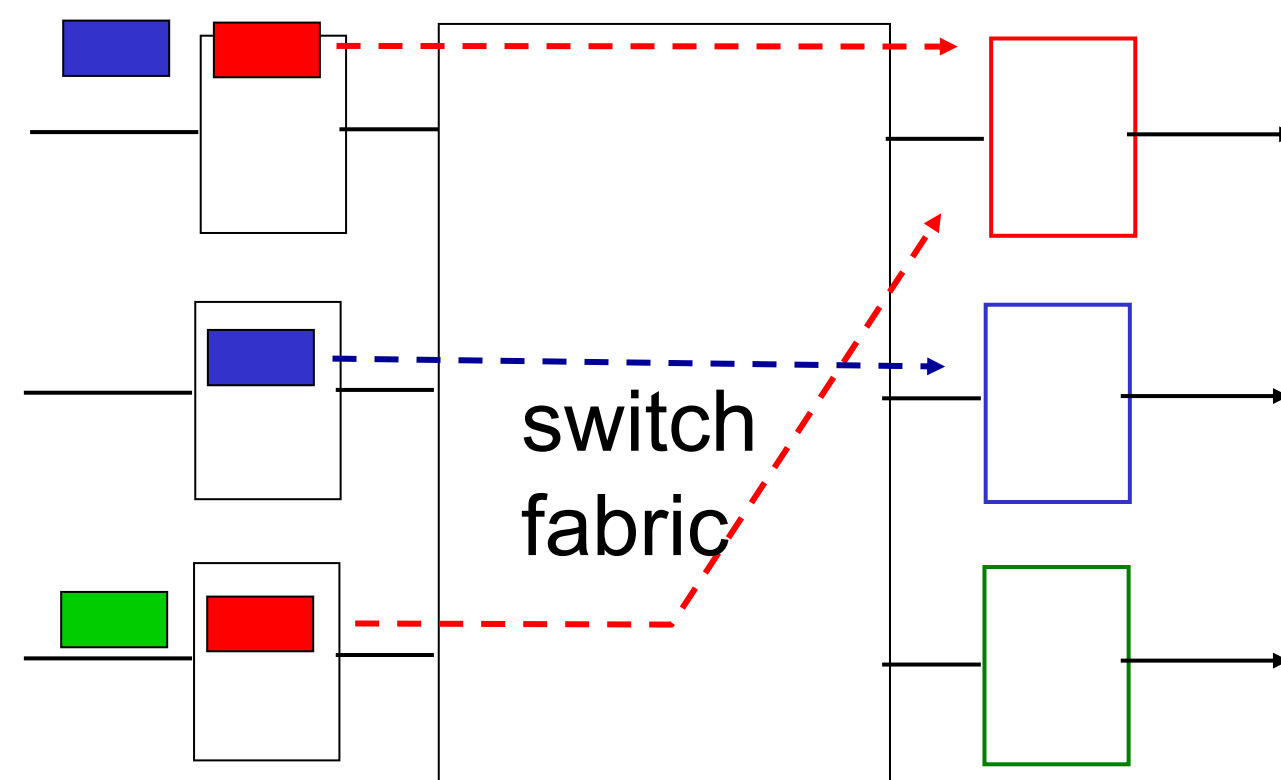
Switching via interconnection network

- ▶ Scaling, using multiple switching “planes” in parallel
 - ▶ Speedup, scaleup via parallelism
- ▶ Cisco CRS router
 - ▶ Basic unit: 8 switching planes
 - ▶ Each plane: 3-stage interconnection network
 - ▶ Up to 100's Tbps switching capacity

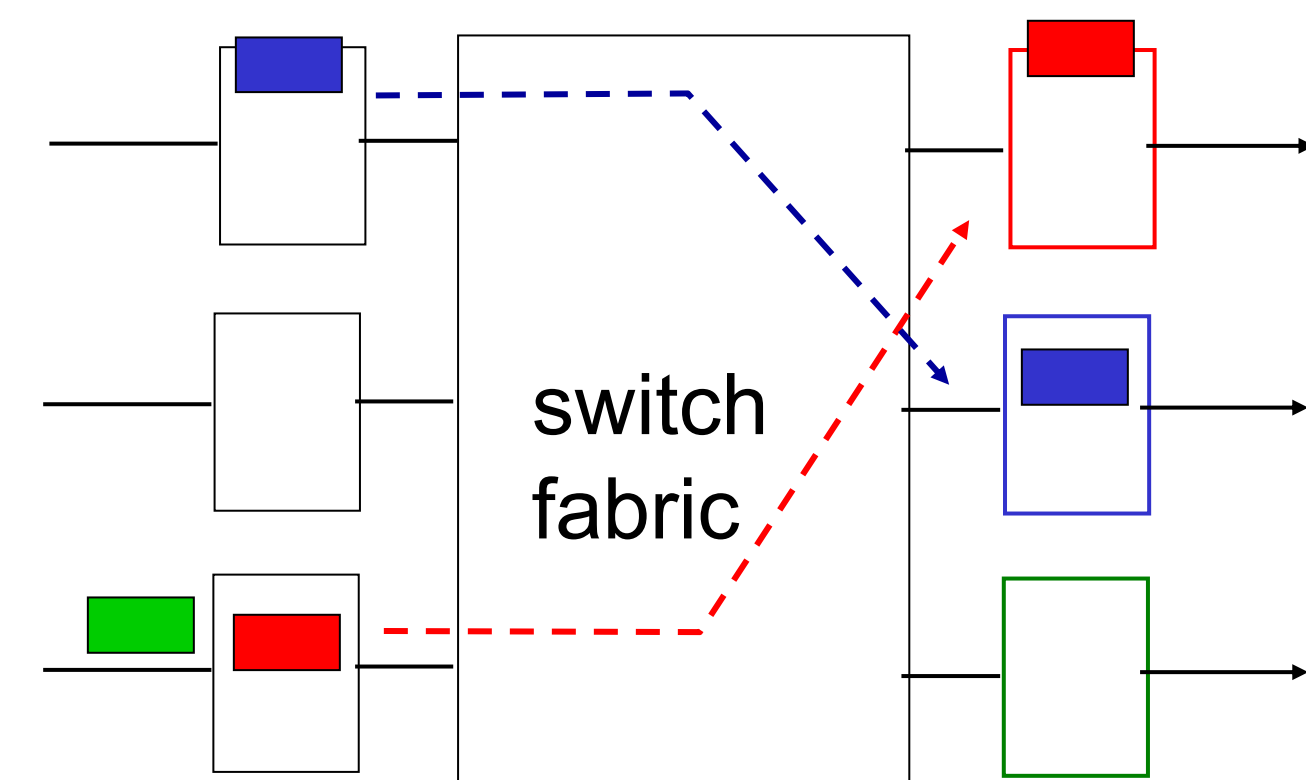


Input port queuing

- ▶ If switch fabric is slower than input ports combined
 - ▶ Queueing may occur at input queues
 - ▶ Queueing delay and loss due to input buffer overflow!
- ▶ Head-of-the-Line (HOL) blocking: queued datagram at front of queue prevents others in queue from moving forward

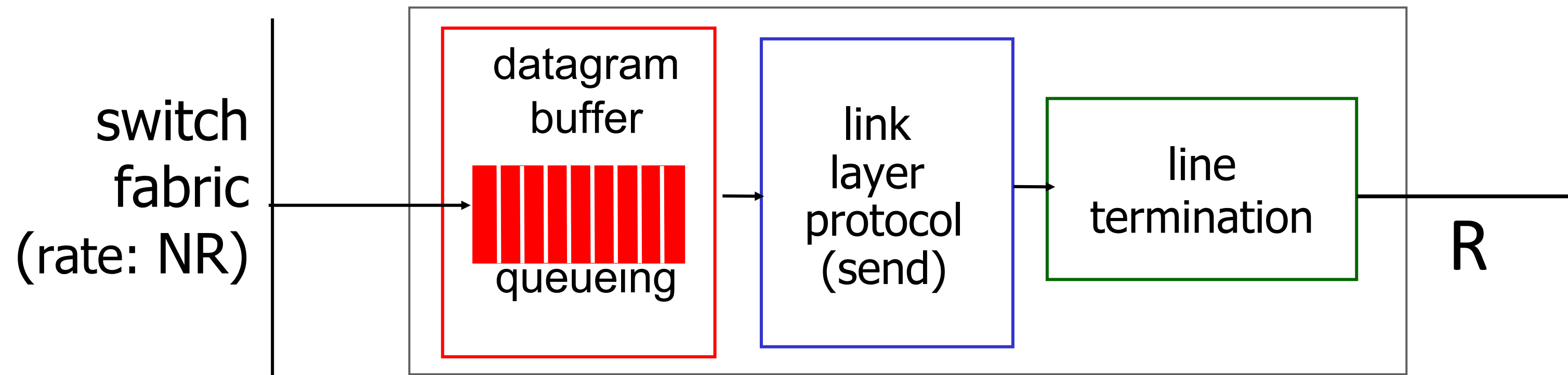


output port contention: only one red datagram can be transferred. lower red packet is *blocked*



one packet time later: green packet experiences HOL blocking

Output port queuing

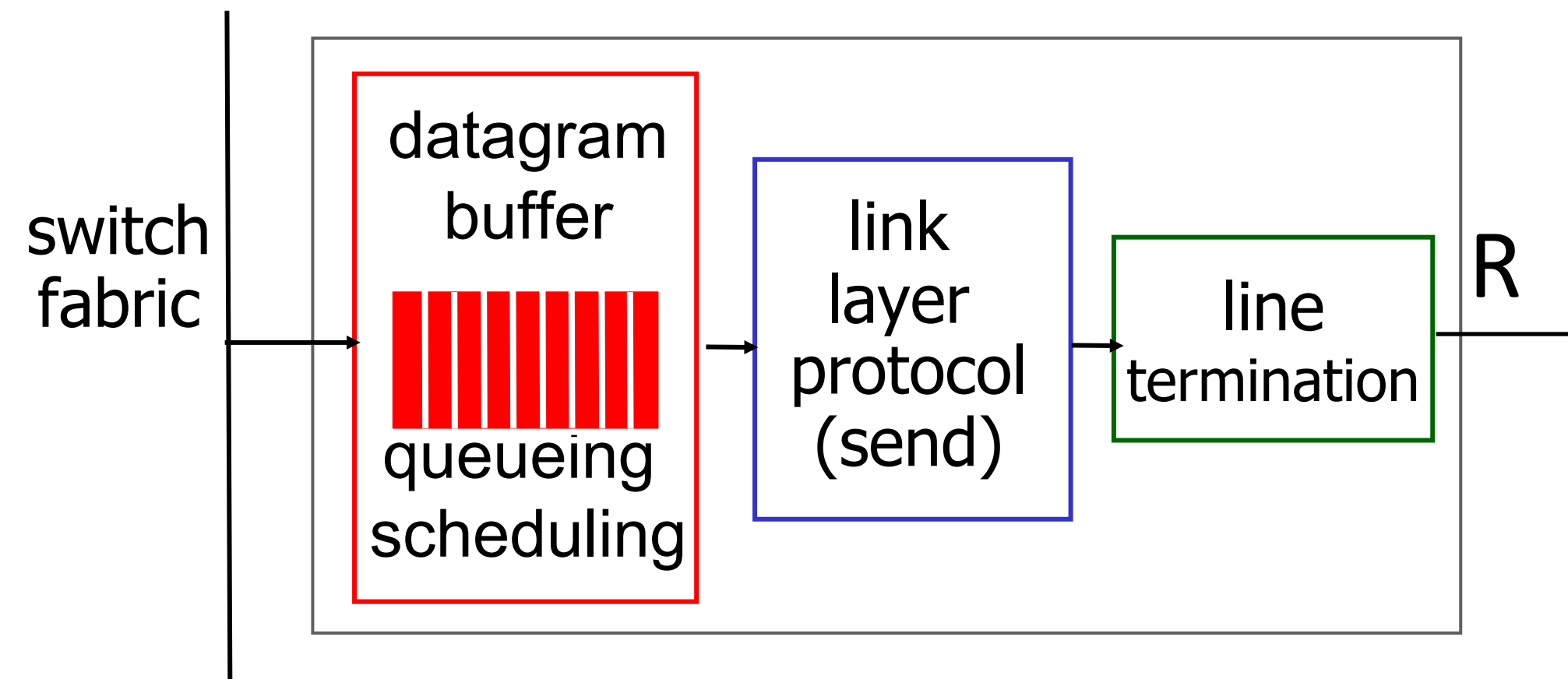


- ▶ Buffering required when datagrams arrive from fabric faster than link transmission rate.
- ▶ **Drop policy:** which datagrams to drop if no free buffers?
- ▶ **Scheduling discipline:** Chooses among queued datagrams for transmission

→ Datagrams can be lost due to congestion, lack of buffers

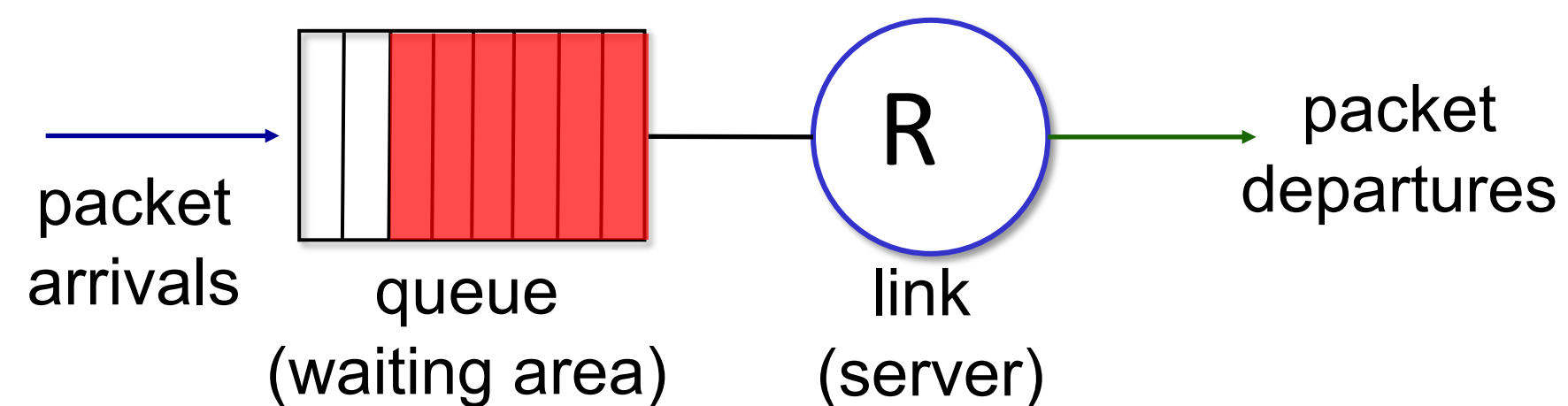
→ Priority scheduling – who gets best performance, network neutrality

Buffer Management



- **Drop:** which packet to add, drop when buffers are full
- **Tail drop:** drop arriving packet
- **Priority:** drop/remove on priority basis
- **Marking:** which packets to mark signal congestion (ECN, RED)

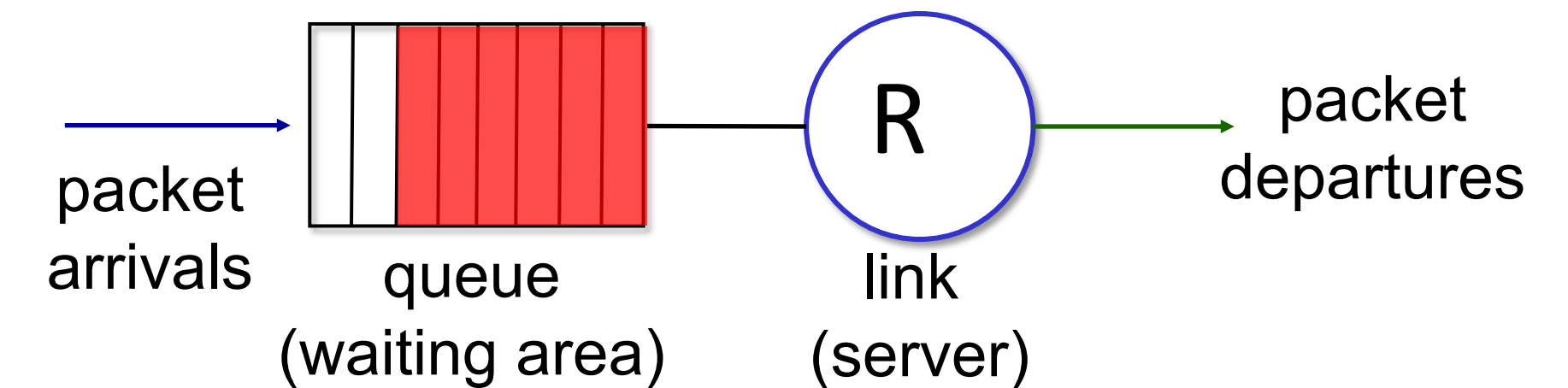
Abstraction: queue



Packet Scheduling – FCFS

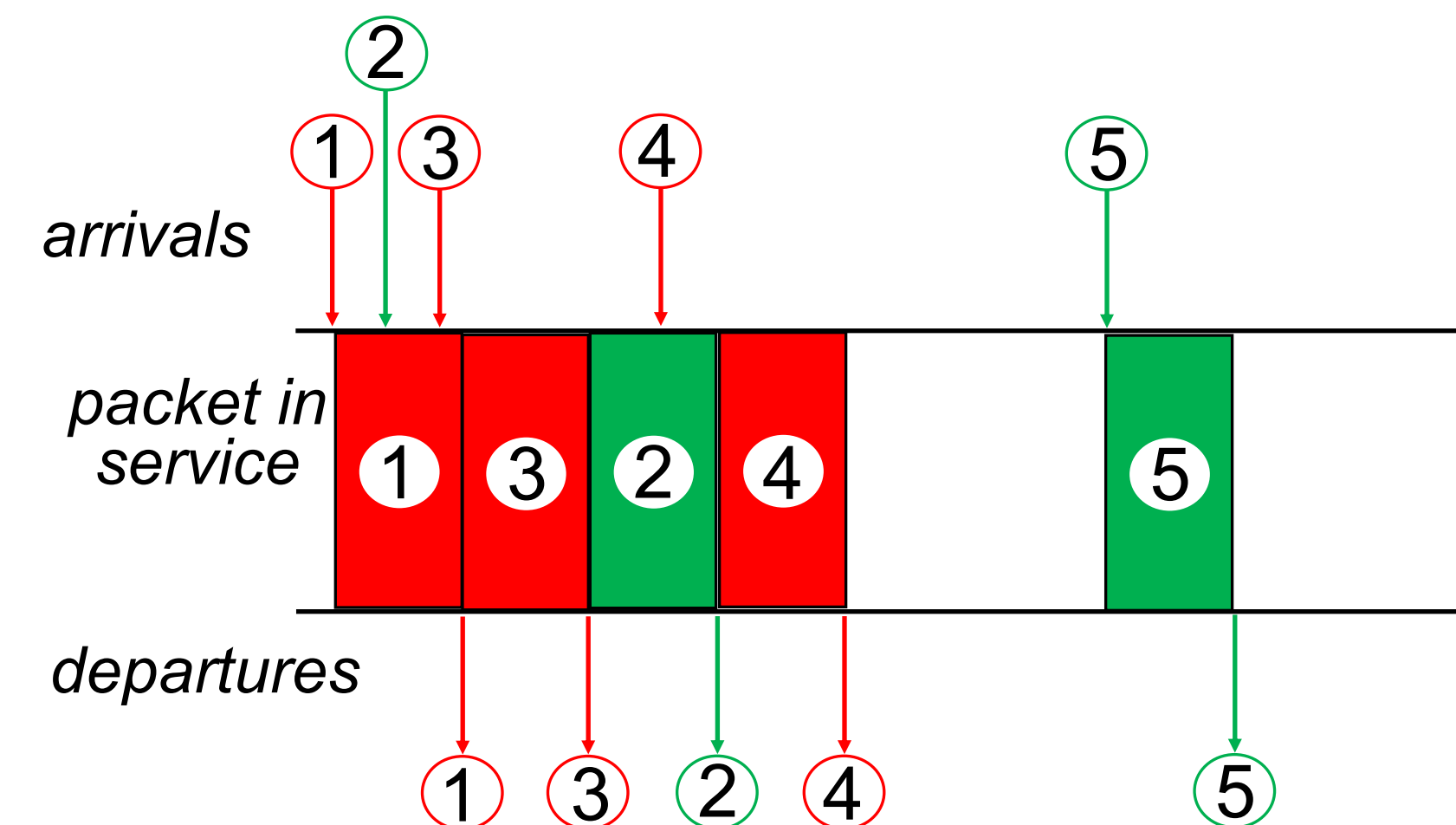
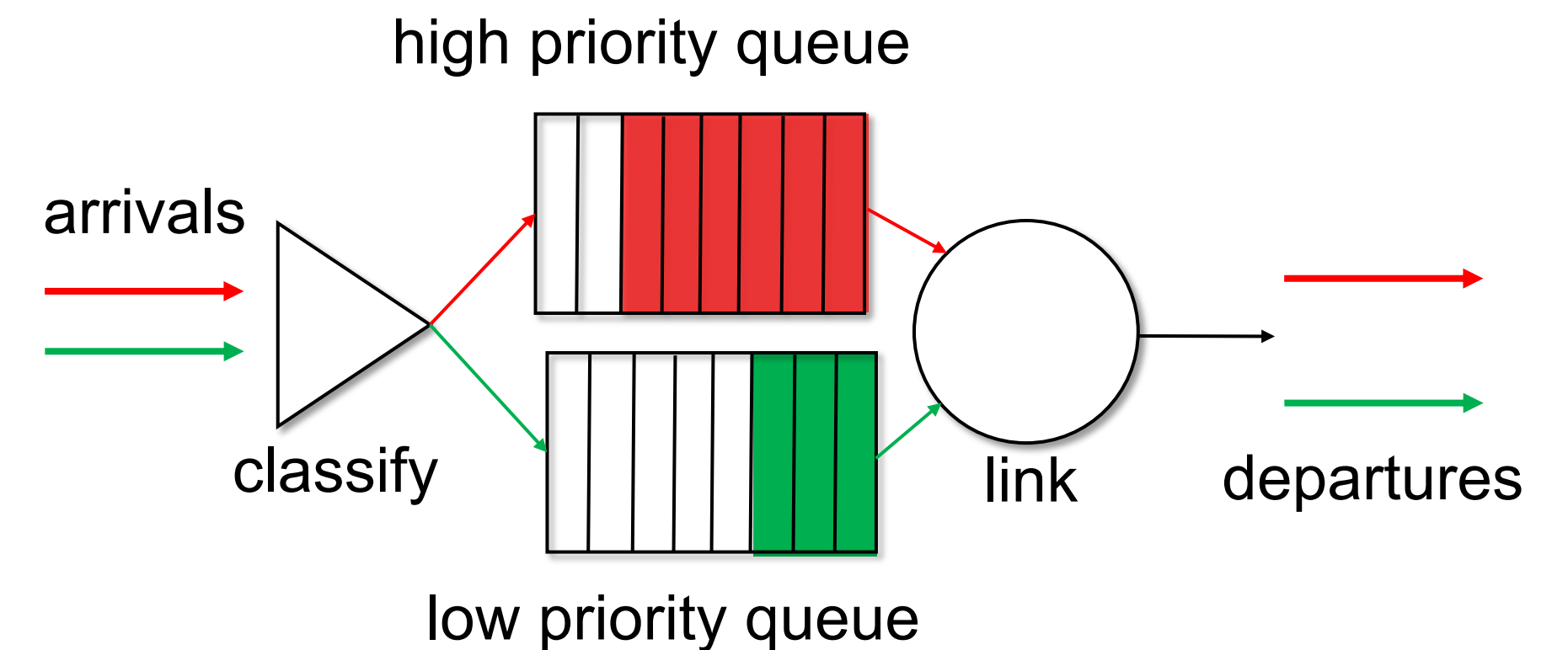
- ▶ **Packet Scheduling:** Deciding which packet to send next on link
- ▶ Priority
- ▶ Round robin
- ▶ Weighted fair queueing
- ▶ First come, first served (FCFS)
 - ▶ Packets transmitted in order of arrival to output port
 - ▶ Also known as: First-in-first-out (FIFO)
 - ▶ Real world example: Checkout at supermarket

Abstraction: queue



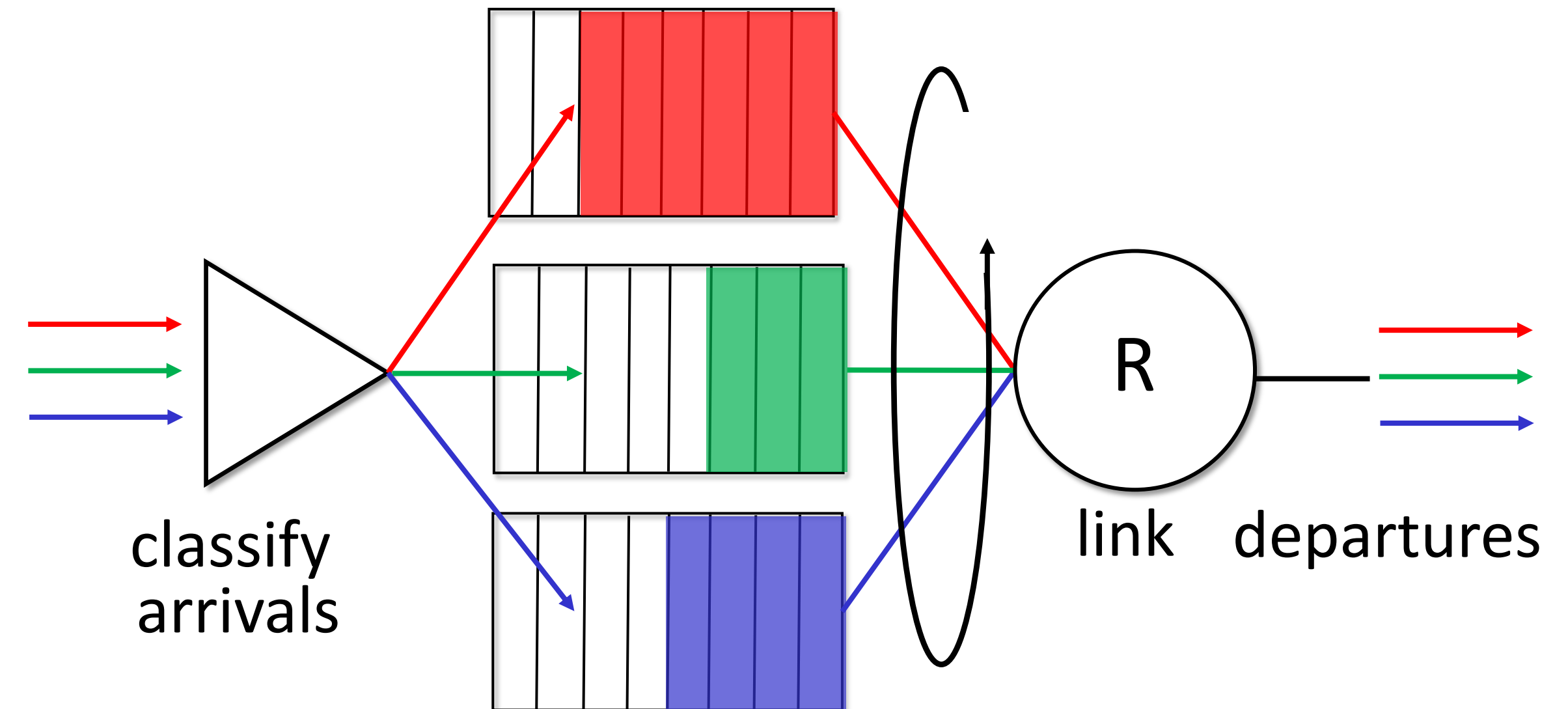
Scheduling policies – Priority Scheduling

- ▶ Arriving traffic classified & queued by class
 - ▶ Any header fields can be used for classification
- ▶ Send packet from highest priority queue that has buffered packets
 - ▶ FCFS within priority class
- ▶ Real world example: Boarding planes



Scheduling policies: Round Robin Scheduling

- ▶ Arriving traffic classified, queued by class
 - ▶ Any header fields can be used for classification
- ▶ Server cyclically, repeatedly scans class queues, sending one complete packet from each class (if available) in turn

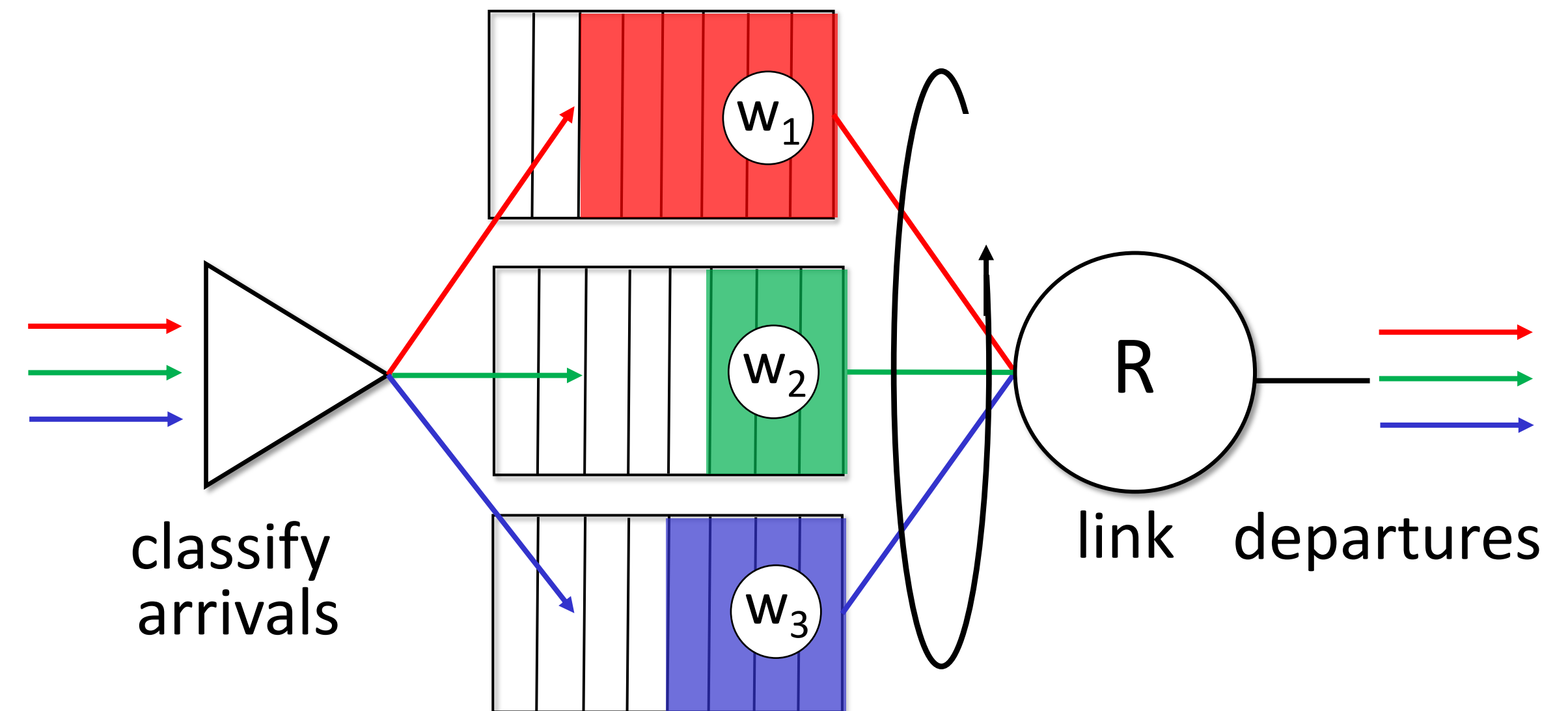


Scheduling policies: Weighted Fair Queueing (WFQ)

- ▶ Generalized Round Robin
- ▶ Each class, i , has weight, w_i , and gets weighted amount of service in each cycle

$$\frac{w_i}{\sum_j w_j}$$

- ▶ Minimum bandwidth guarantee (per-traffic-class)



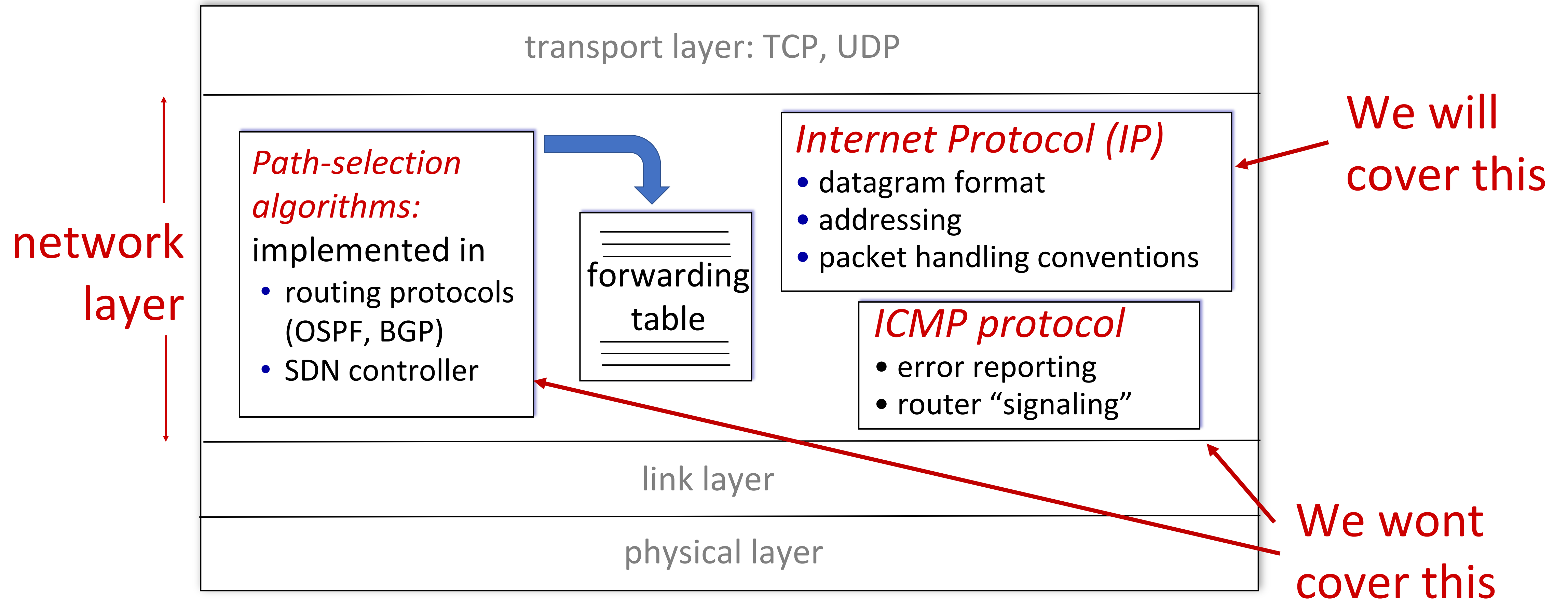
Interlude: Network Neutrality

- ▶ What is network neutrality?
- ▶ Technical: how an ISP should share/allocate its resources
 - ▶ Packet scheduling, buffer management are the mechanisms
- ▶ Social, economic principles
 - ▶ Protecting free speech
 - ▶ Encouraging innovation, competition
 - ▶ Enforced legal rules and policies

Interlude: Network Neutrality

- ▶ How does Network Neutrality affect you?
- ▶ Many mobile data providers offered zero-rating options
 - ▶ E.g., streaming services like Netflix, Spotify etc. were not counted towards the data allowance in the contract
- ▶ Court of Justice of the European Union Decision of 2nd sept 2021
 - ▶ This is not allowed due to net neutrality
 - ▶ Smaller, upcoming streaming services would not be included in the zero-rating and therefore have a systematic disadvantage
 - ▶ Operators had to stop offering these options and phase them out even for existing customers

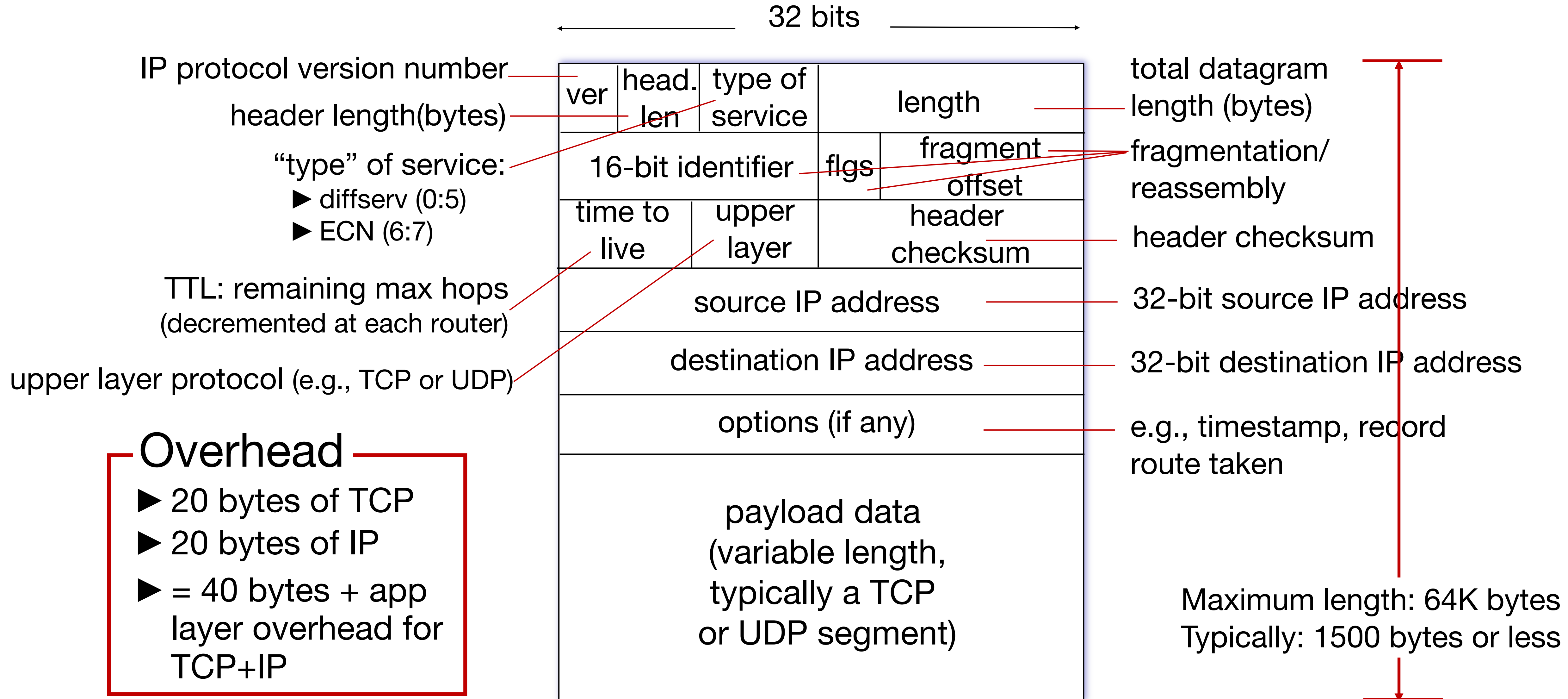
Network Layer: Internet



Learning Goal 7A5281

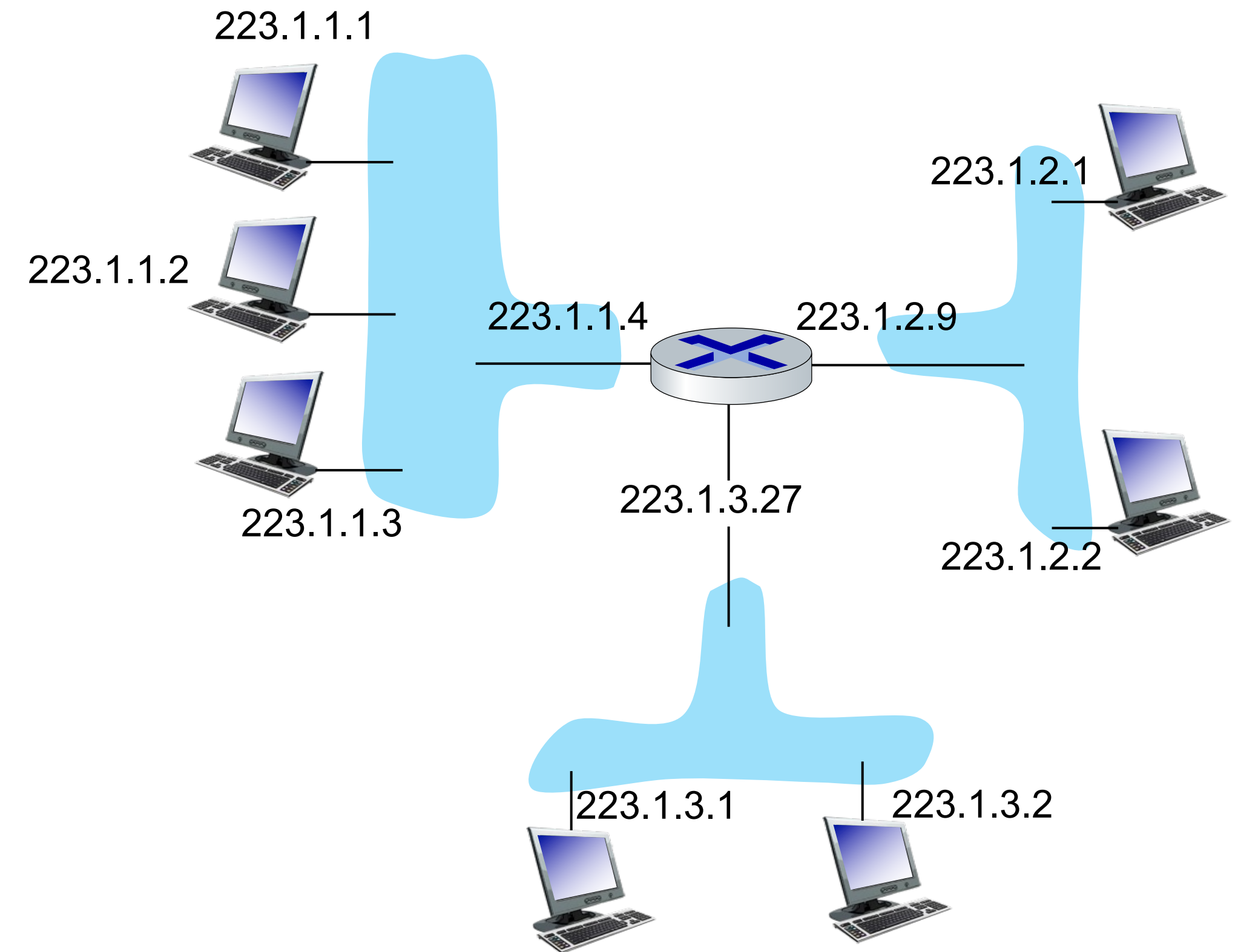
**What is the Internet Protocol (v4) and
how does IP addressing work?**

IPv4 Datagram format



IP addressing – Introduction

- ▶ **IP address:** 32-bit identifier associated with each host or router interface
- ▶ **Interface:** connection between host/router and physical link
- ▶ Router's typically have multiple interfaces
- ▶ Host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)



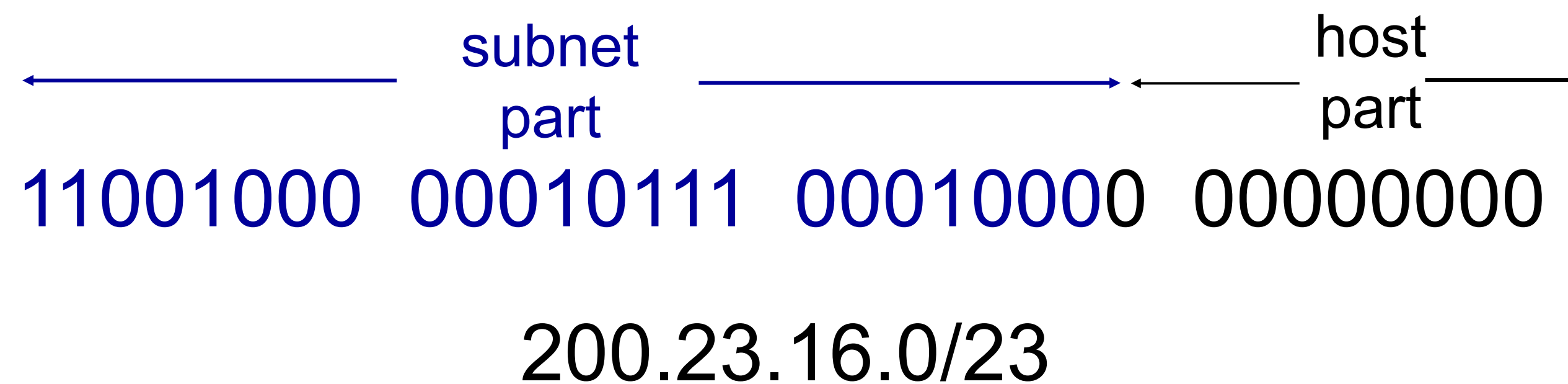
dotted-decimal IP address notation:

223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

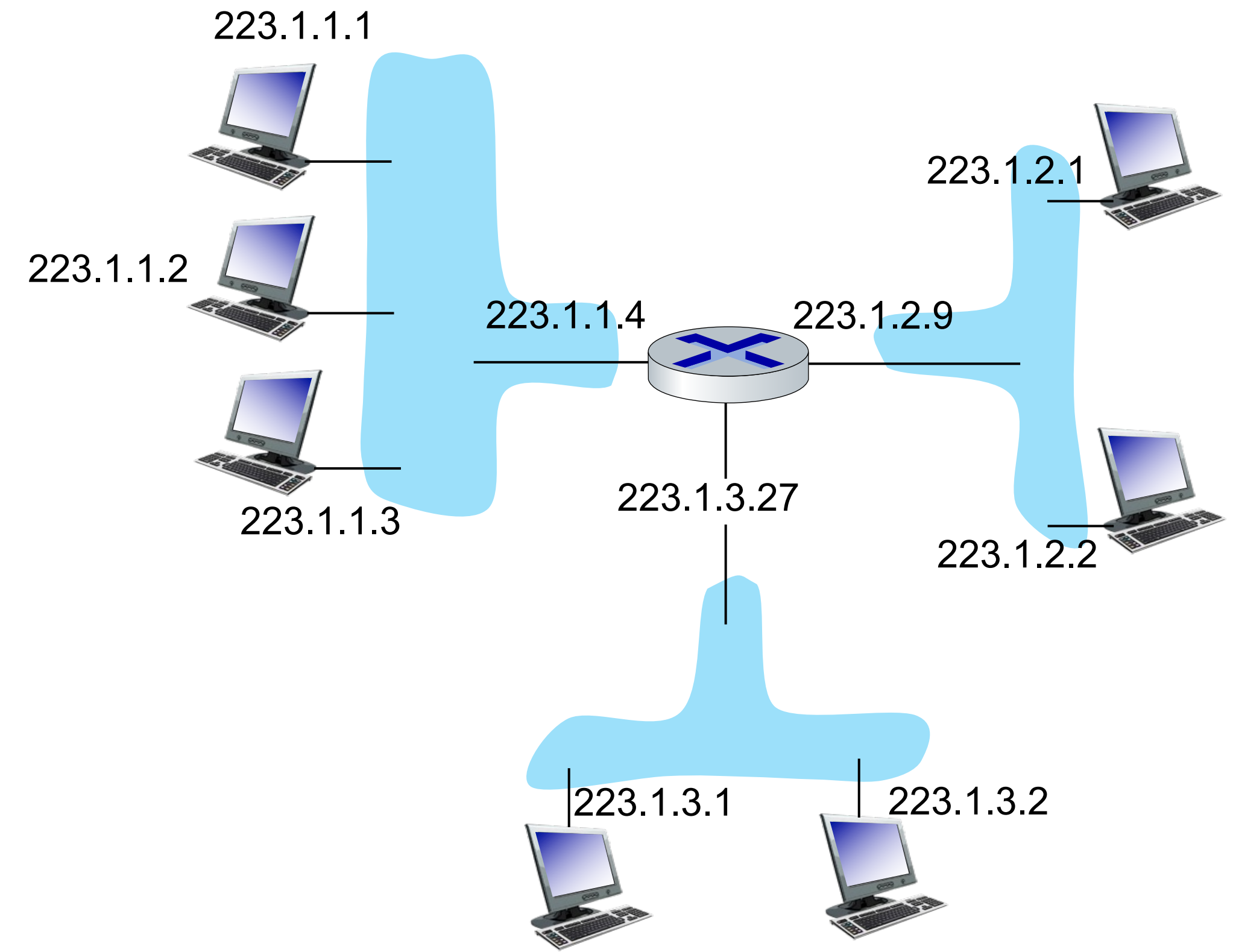
IP addressing – CIDR

- ▶ CIDR: Classless InterDomain Routing (pronounced “cider”)
 - ▶ Subnet portion of address of arbitrary length
 - ▶ Address format: a.b.c.d/x, where x is # bits in subnet portion of address



IP addressing – Subnets

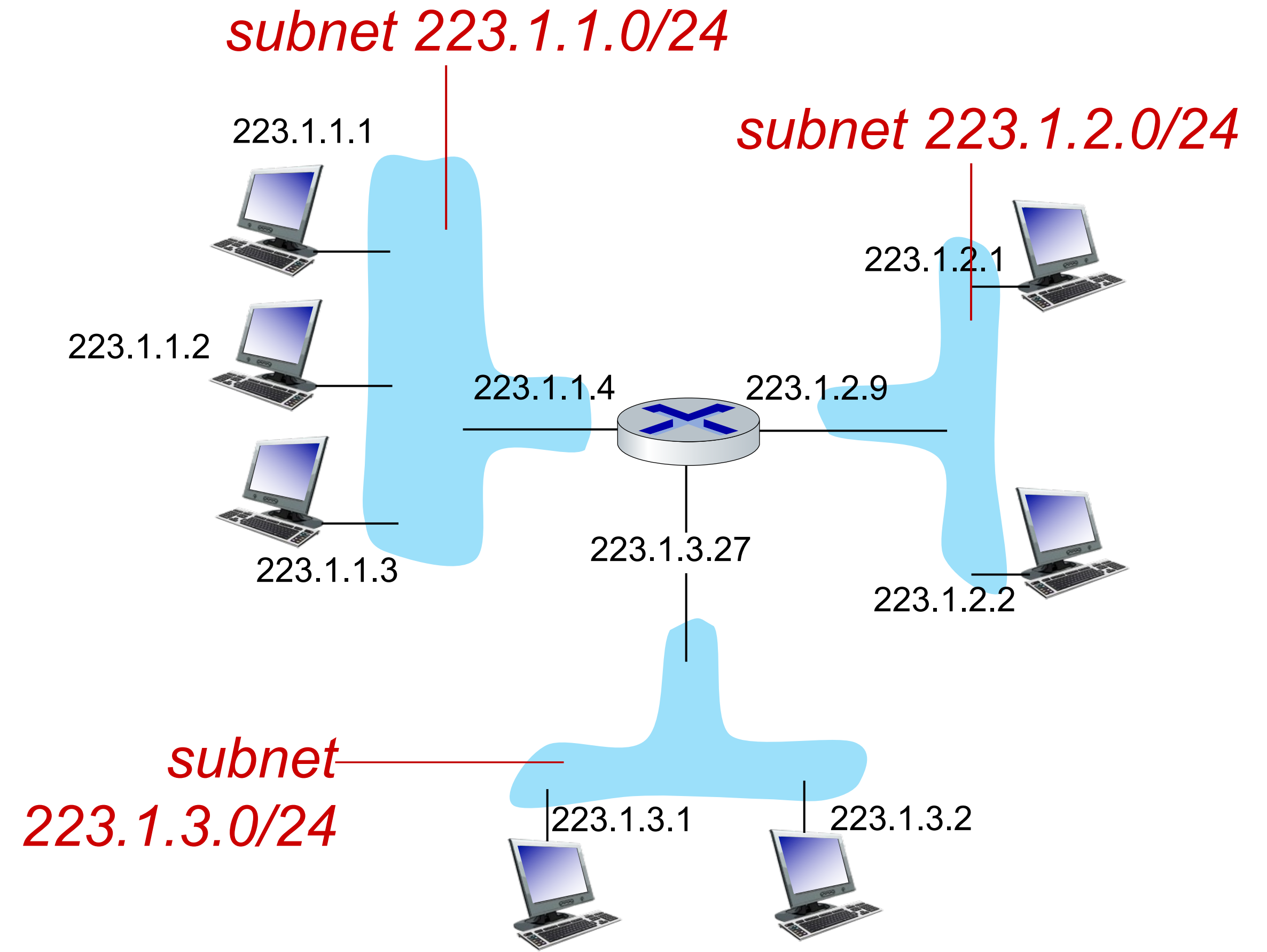
- ▶ What's a subnet ?
 - ▶ Device interfaces that can physically reach each other without passing through an intervening router
- ▶ IP addresses have structure
 - ▶ Subnet part: devices in same subnet have common high order bits
 - ▶ Host part: remaining low order bits



network consisting of 3 subnets

IP addressing – Subnets

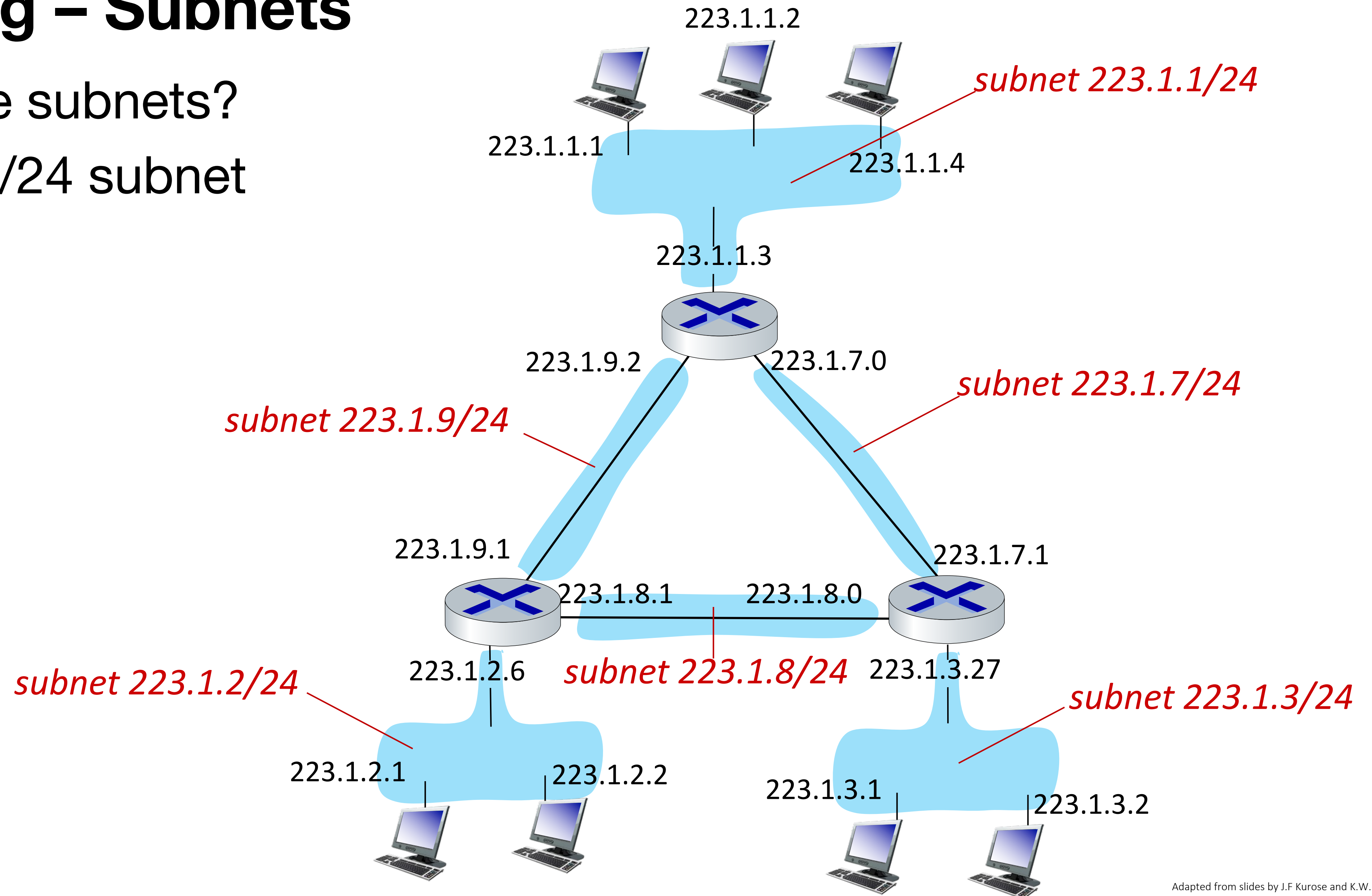
- ▶ Recipe for defining subnets
 - ▶ Detach each interface from its host or router, creating “islands” of isolated networks
 - ▶ Each isolated network is called a subnet



subnet mask: /24
(high-order 24 bits: subnet part
of IP address)

IP addressing – Subnets

- Where are the subnets?
- What are the /24 subnet addresses?



IP addressing – How to get one

- ▶ That's actually two questions:

- a) Q: How does a host get IP address within its network (host part of address)?
- b) Q: How does a network get IP address for itself (network part of address)?

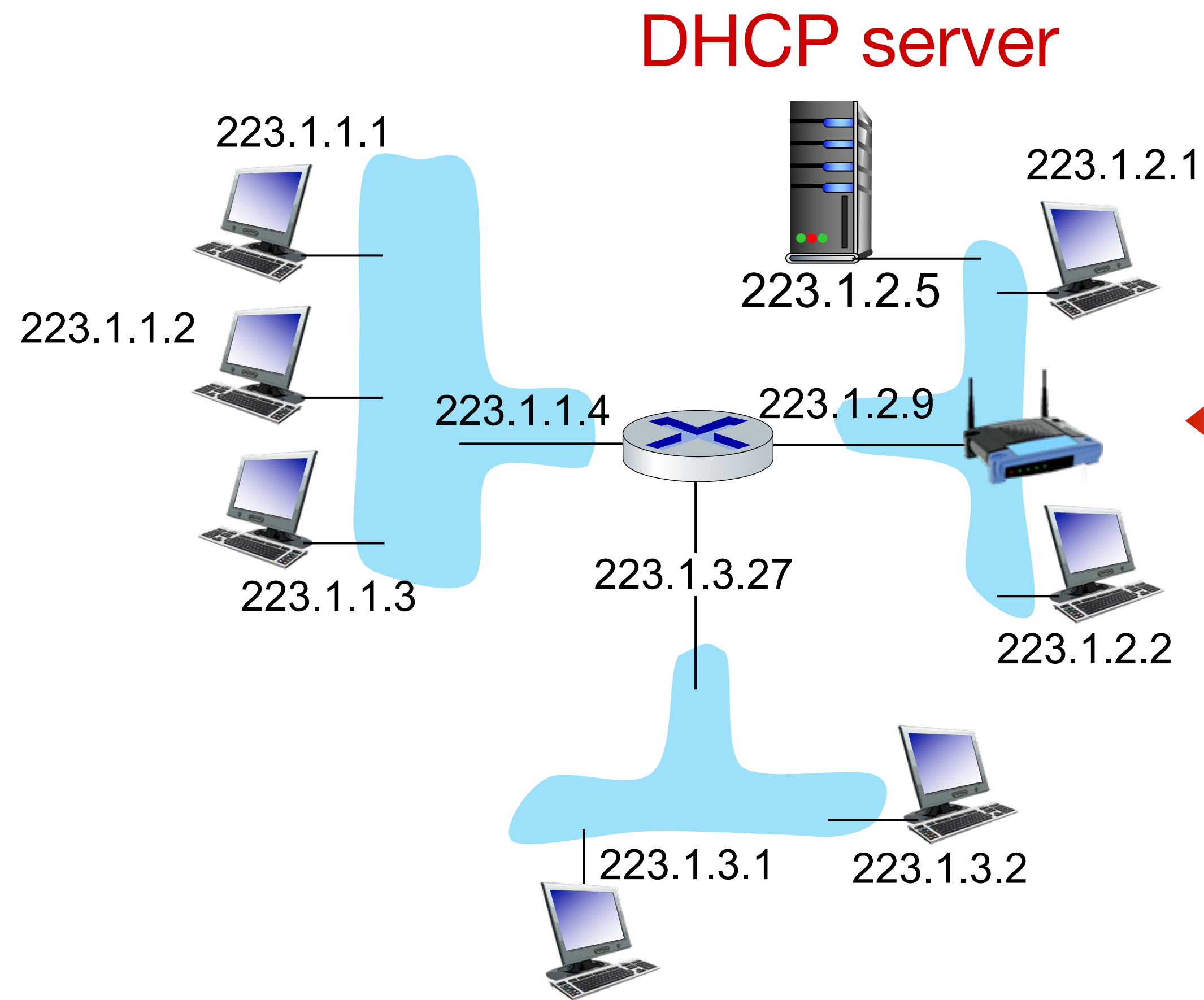
- ▶ How does host get IP address?

- ▶ Hard-coded by sysadmin in config file (e.g., /etc/rc.config in UNIX)
- ▶ DHCP: Dynamic Host Configuration Protocol
 - ▶ Dynamically get address from as server
 - ▶ “Plug-and-play” → zeroconf

Dynamic Host Configuration Protocol (DHCP)

- ▶ **Goal:** Host dynamically obtains IP address from network server when it “joins” network
 - ▶ Can renew its lease on address in use
 - ▶ Allows reuse of addresses (only hold address while connected/on)
 - ▶ Support for mobile users who join/leave network
- ▶ DHCP overview
 - ▶ Host broadcasts DHCP discover msg [optional]
 - ▶ DHCP server responds with DHCP offer msg [optional]
 - ▶ Host requests IP address: DHCP request msg
 - ▶ DHCP server sends address: DHCP ack msg

DHCP client-server scenario



Typically, DHCP server will be co-located in router, serving all subnets to which router is attached



arriving **DHCP client** needs address in this network

DHCP client-server scenario

DHCP server: 223.1.2.5

DHCP discover

Broadcast: is there a
DHCP server out there?

Arriving client

DHCP offer

Broadcast: I'm a DHCP
server! Here's an IP
address you can use

DHCP request

Broadcast: OK. I would like
to use this IP address!

DHCP ACK

Broadcast: OK. You've
got that IP address!

These two steps can
be skipped "if a client
remembers and
wishes to reuse a
previously allocated
network address"
[RFC 2131]

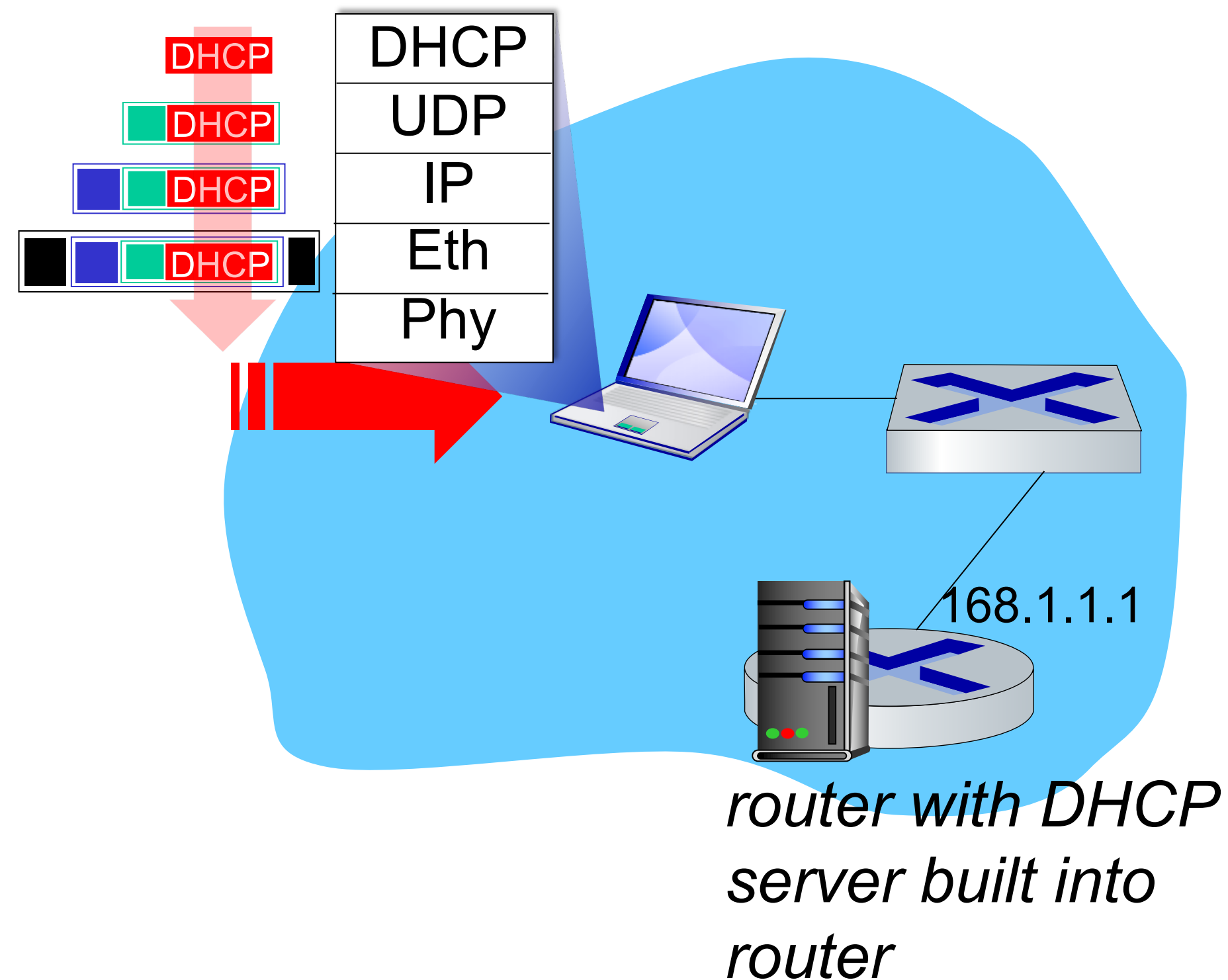
Q: Why do we use a
broadcast in the last two
steps here?

A: There might be
multiple DHCP servers

DHCP: More than IP addresses

- ▶ DHCP can return more than just allocated IP address on subnet
 - ▶ Address of first-hop router for client
 - ▶ Name and IP address of DNS sever
 - ▶ Network mask (indicating network versus host portion of address)

DHCP: Example

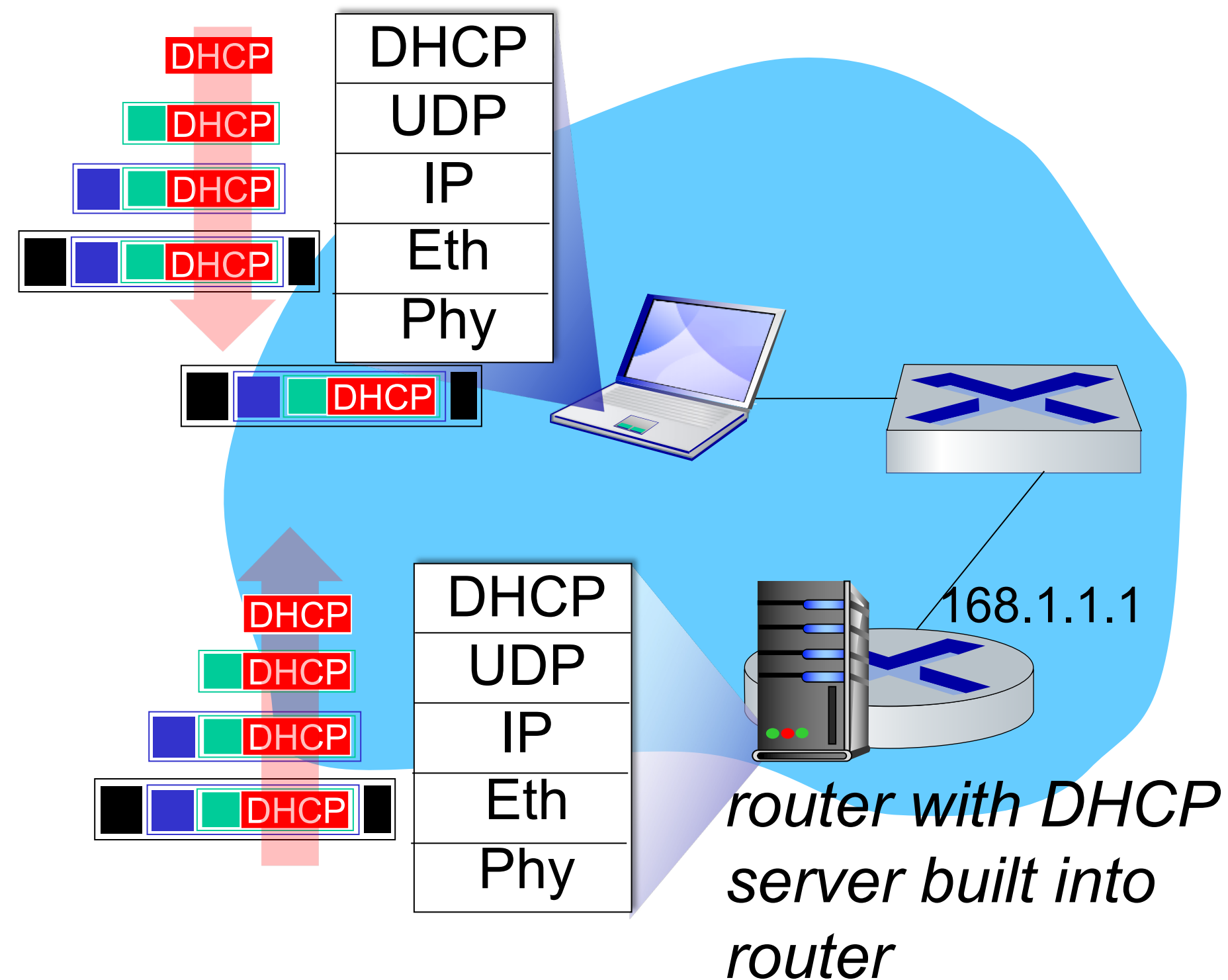


- ▶ Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server
- ▶ DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet

Q: Why is DHCP using UDP as a transport layer protocol?

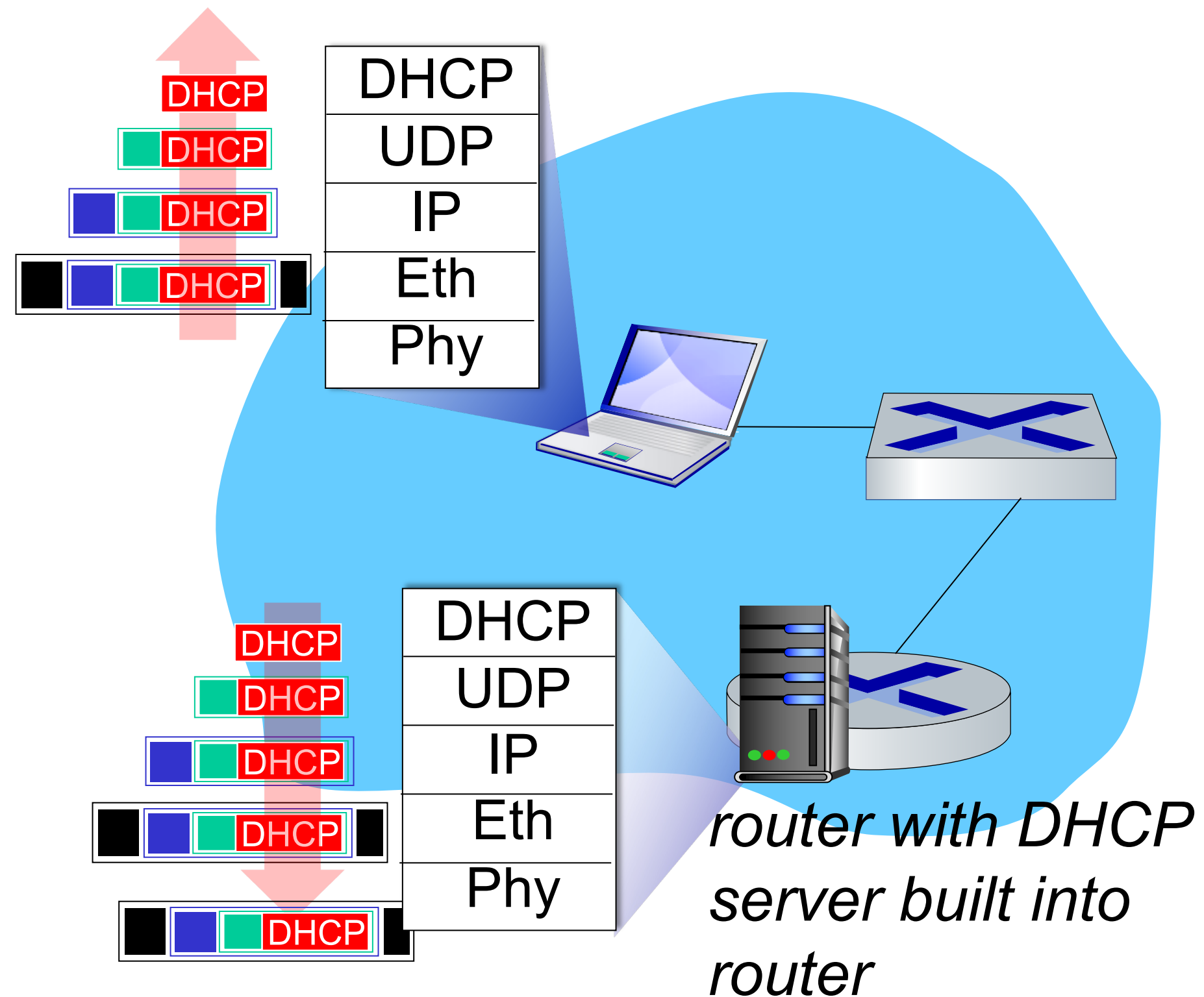
A: Without address we cannot perform a TCP handshake

DHCP: Example



- ▶ Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server
- ▶ DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet
- ▶ Ethernet frame broadcast (dest: FFFFFFFFFFFFFFFF) on LAN, received at router running DHCP server
- ▶ Ethernet de-mux'ed to IP de-mux'ed, UDP de-mux'ed to DHCP

DHCP: Example



- ▶ DHCP server formulates DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- ▶ Encapsulated DHCP server reply forwarded to client, de-muxing up to DHCP at client
- ▶ Client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

How does a network get the subnet part of IP address?

- Gets allocated portion of its provider ISP's address space

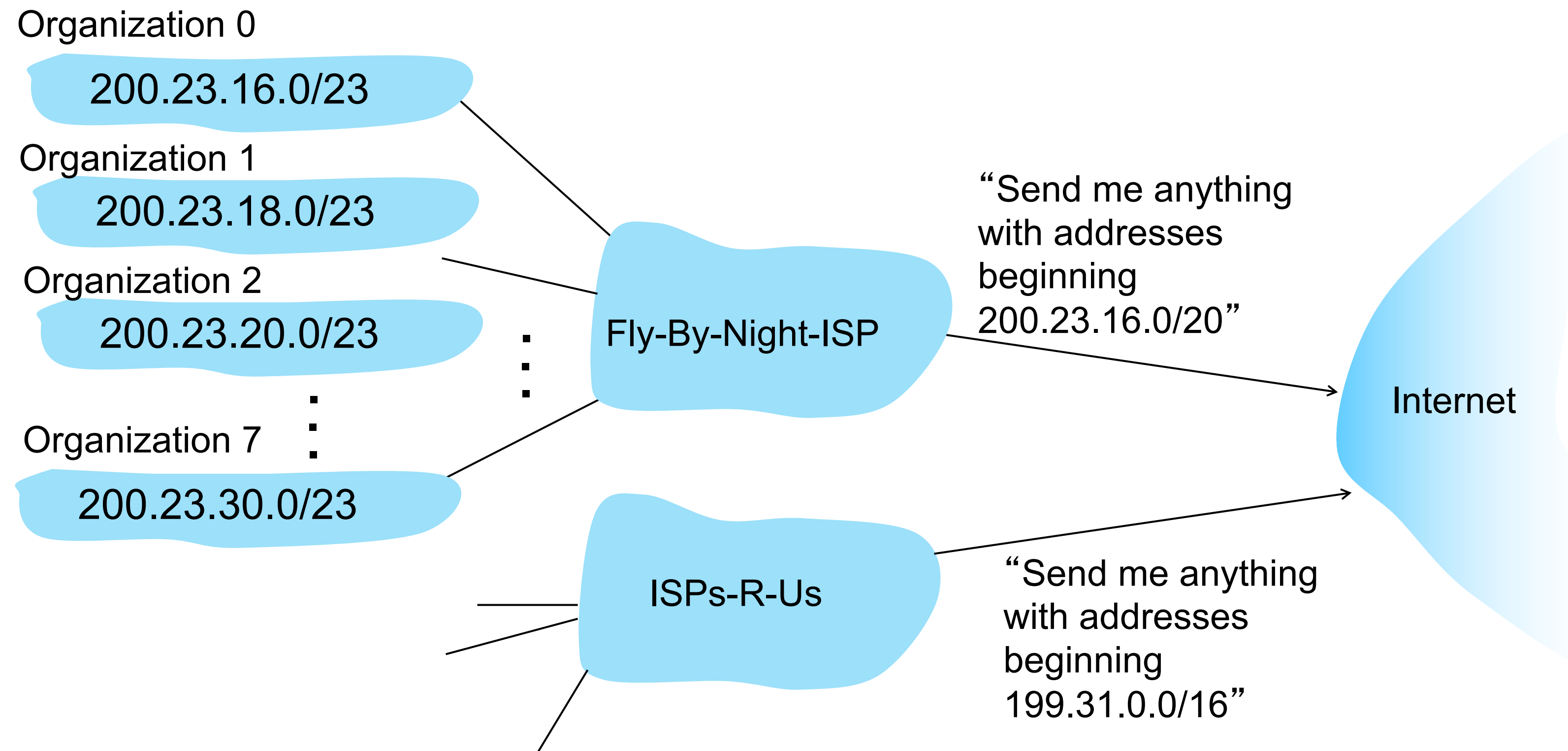
ISP's block 11001000 00010111 00010000 00000000 200.23.16.0/20

ISP can then allocate out its address space in blocks:

Organization 0	<u>11001000 00010111 00010000</u>	00000000	200.23.16.0/23
Organization 1	<u>11001000 00010111 00010010</u>	00000000	200.23.18.0/23
Organization 2	<u>11001000 00010111 00010100</u>	00000000	200.23.20.0/23
...			
Organization 7	<u>11001000 00010111 00011110</u>	00000000	200.23.30.0/23

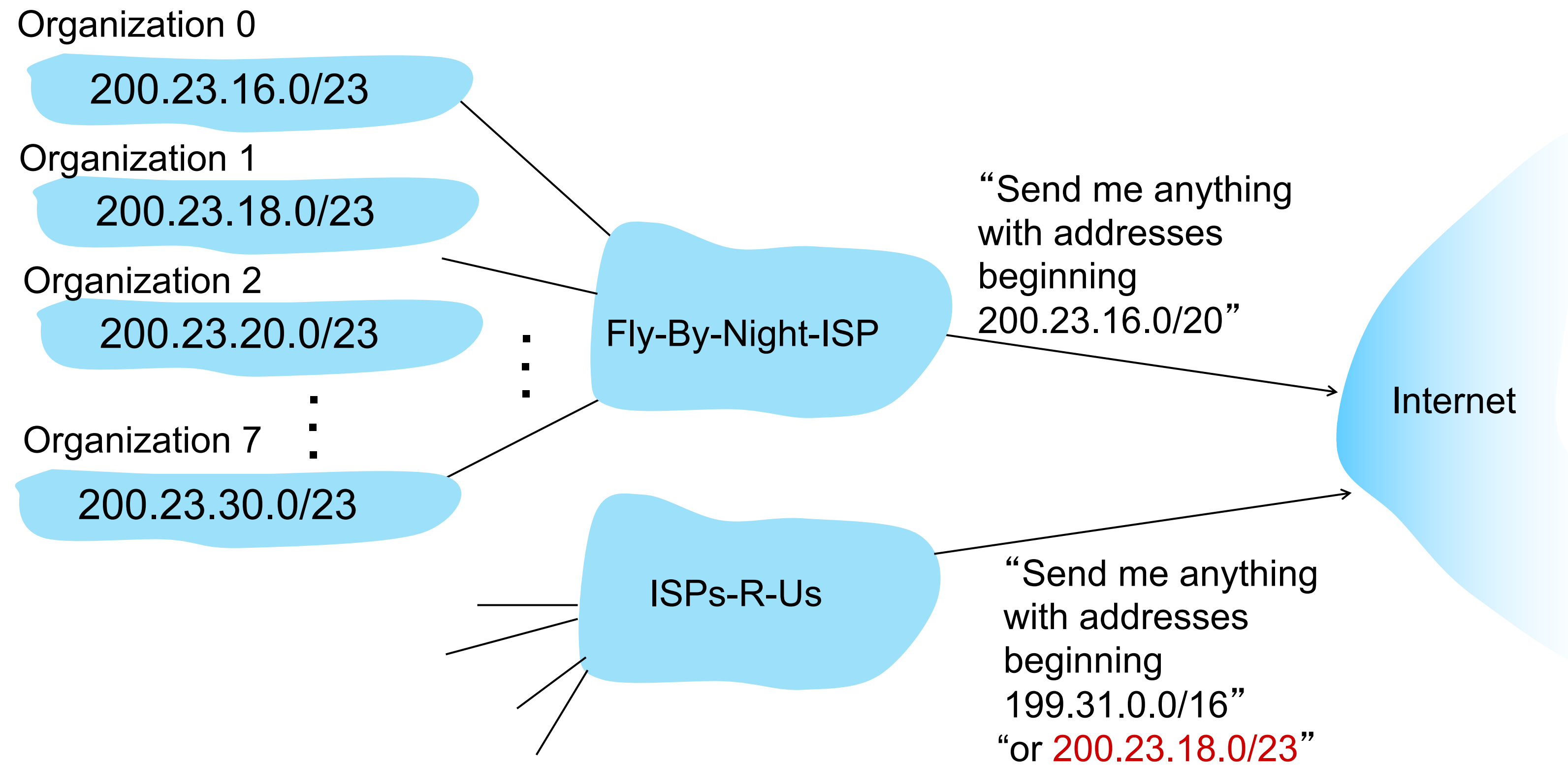
Hierarchical addressing: route aggregation

- Hierarchical addressing allows efficient advertisement of routing information



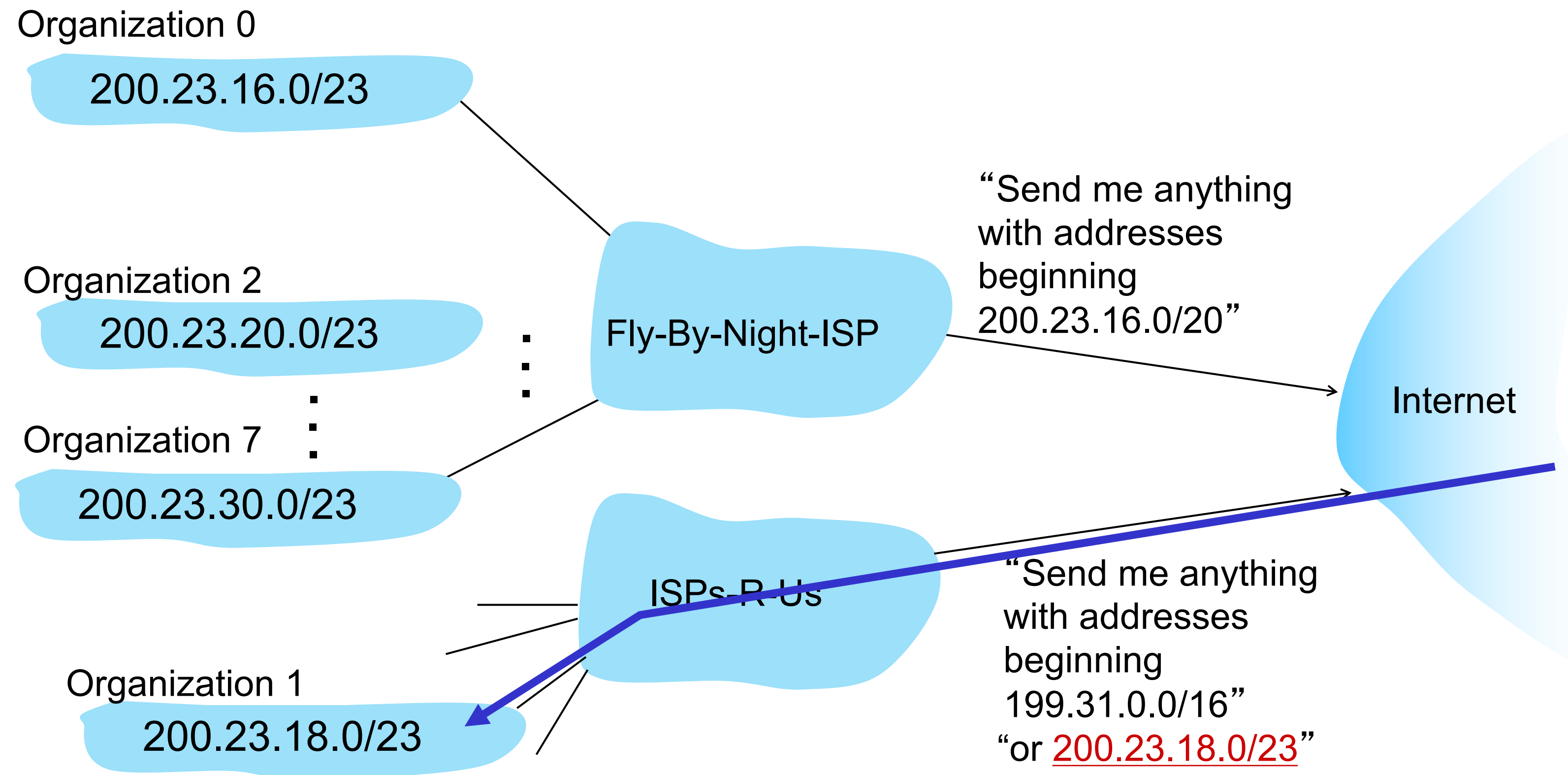
Hierarchical addressing: route aggregation

- Organization 1 moves from Fly-By-Night-ISP to ISPs-R-Us
- ISPs-R-Us now advertises a more specific route to Organization 1



Hierarchical addressing: route aggregation

- Organization 1 moves from Fly-By-Night-ISP to ISPs-R-Us
- ISPs-R-Us now advertises a more specific route to Organization 1



IP addressing: last words ...

- ▶ **Q:** How does an ISP get block of addresses?
- ▶ **A:** ICANN: Internet Corporation for Assigned Names and Numbers <http://www.icann.org/>
 - ▶ Allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
 - ▶ Manages DNS root zone, including delegation of individual TLD (.com, .edu , ...) management

- ▶ **Q:** are there enough 32-bit IP addresses?
- ▶ ICANN allocated last chunk of IPv4 addresses in 2011
- ▶ NAT (next up) helps IPv4 address space exhaustion
- ▶ IPv6 has 128-bit address space

"Who the hell knew how much address space we needed?"

- Vint Cerf

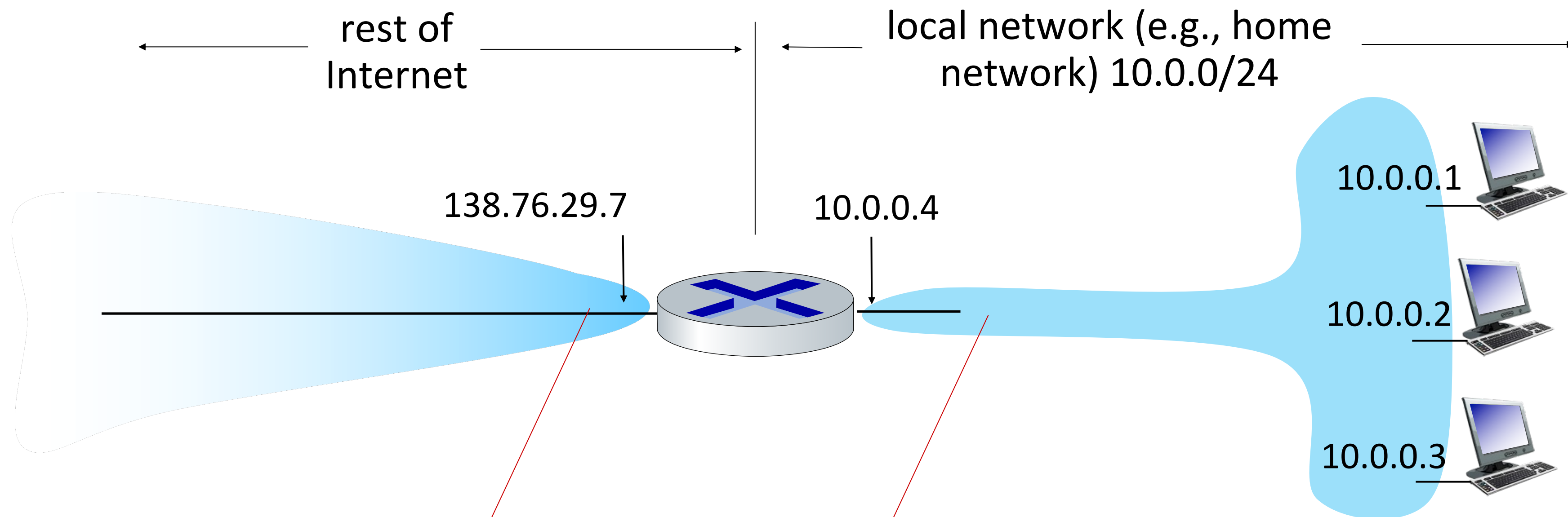
(reflecting on decision to make IPv4 address 32 bits long)

Learning Goal 63B1FD

What is Network Address Translation?

Network Address Translation (NAT)

- All devices in local network share just one IPv4 address as far as outside world is concerned



all datagrams *leaving* local network have *same* source NAT IP address: 138.76.29.7, but *different* source port numbers

datagrams with source or destination in this network have 10.0.0/24 address for source, destination (as usual)

Network Address Translation (NAT)

- ▶ All devices in local network have 32-bit addresses in a “private” IP address space (10/8, 172.16/12, 192.168/16 prefixes) that can only be used in local network
- ▶ Advantages
 - ▶ Just one IP address needed from provider ISP for all devices
 - ▶ Can change addresses of host in local network without notifying outside world
 - ▶ Can change ISP without changing addresses of devices in local network
 - ▶ Security: devices inside local net not directly addressable/visible by outside world

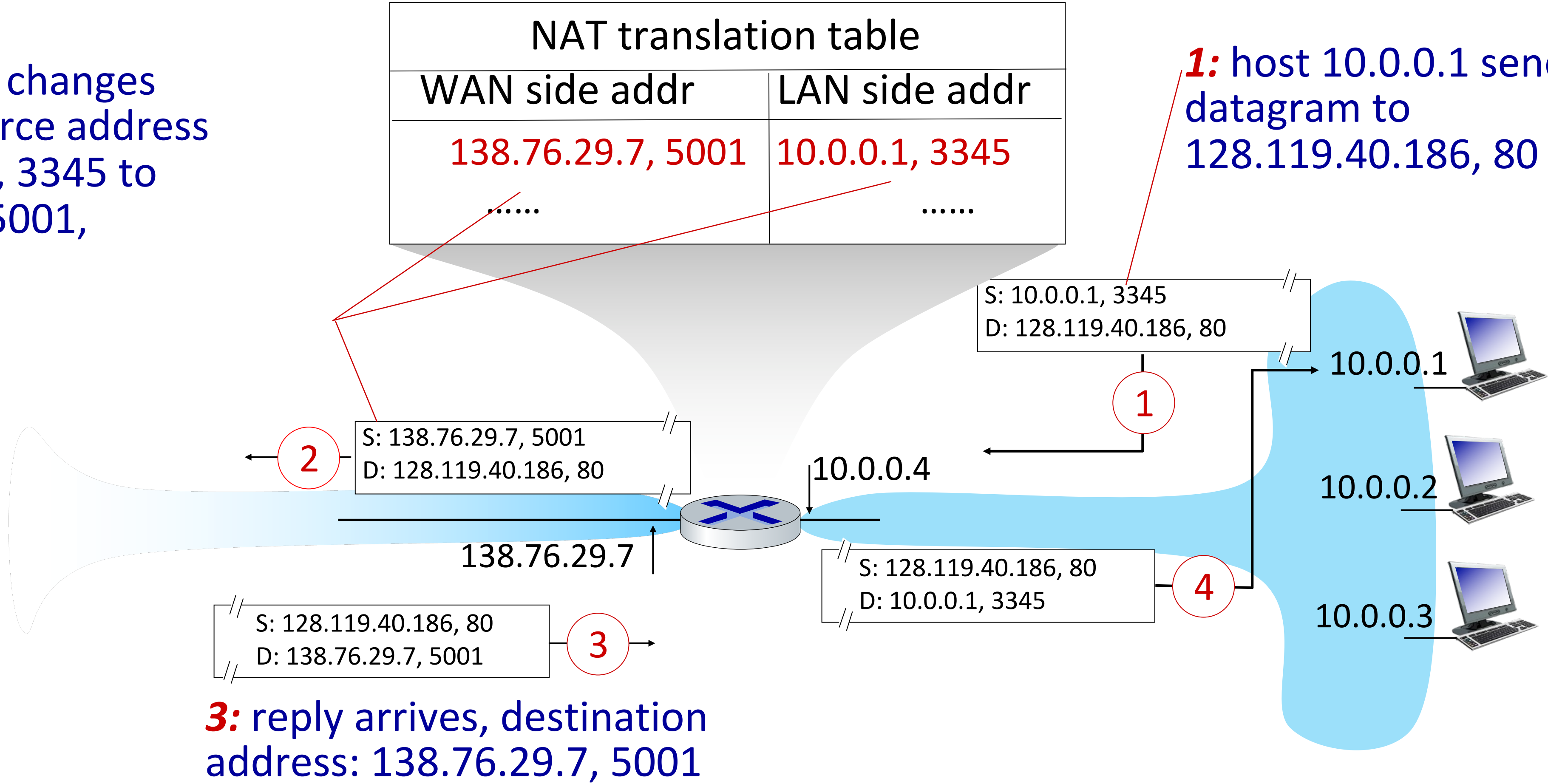
NAT – Implementation

- ▶ NAT router must (transparently)
 - ▶ Outgoing datagrams: replace (source IP address, port #) of every outgoing datagram to (NAT IP address, new port #)
 - ▶ Remote clients/servers will respond using (NAT IP address, new port #) as destination address
 - ▶ Remember (in NAT translation table) every (source IP address, port #) to (NAT IP address, new port #) translation pair
 - ▶ Incoming datagrams: replace (NAT IP address, new port #) in destination fields of every incoming datagram with corresponding (source IP address, port #) stored in NAT table

NAT – Illustrated

2: NAT router changes datagram source address from 10.0.0.1, 3345 to 138.76.29.7, 5001, updates table

1: host 10.0.0.1 sends datagram to 128.119.40.186, 80



NAT - Controversies

- ▶ NAT has been controversial...
 - ▶ Routers “should” only process up to layer 3
 - ▶ Violates end-to-end argument (port # manipulation by network-layer device)
 - ▶ Address “shortage” should be solved by IPv6
 - ▶ NAT traversal: what if client wants to connect to server behind NAT?
- ▶ ...but NAT is here to stay:
 - ▶ Extensively used in home and institutional nets, 4G/5G cellular nets

Learning Goal 5F7391

What is the Internet Protocol v6?

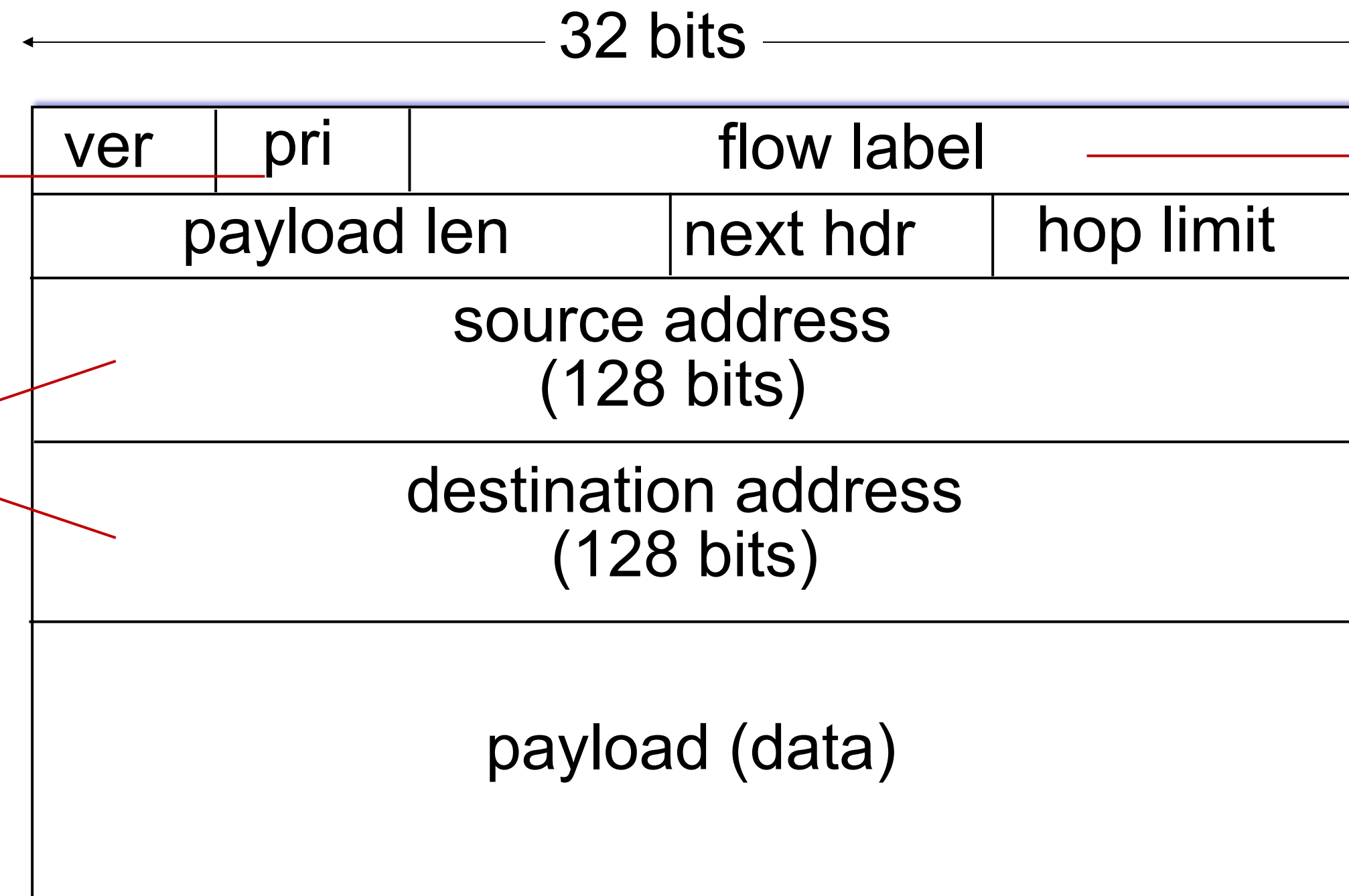
Internet Protocol version 6 (IPv6)

- ▶ Initial motivation: 32-bit IPv4 address space completely allocated
- ▶ Additional motivation
 - ▶ Speed processing/forwarding: 40-byte fixed length header
 - ▶ Enable different network-layer treatment of “flows”
- ▶ Sidenote: IPv5 was “skipped” since it carried over some limitations of IPv4, especially 32-bit addresses

IPv6 – Datagram Format

Priority: identify
priority among
datagrams in flow

128-bit
IPv6 addresses



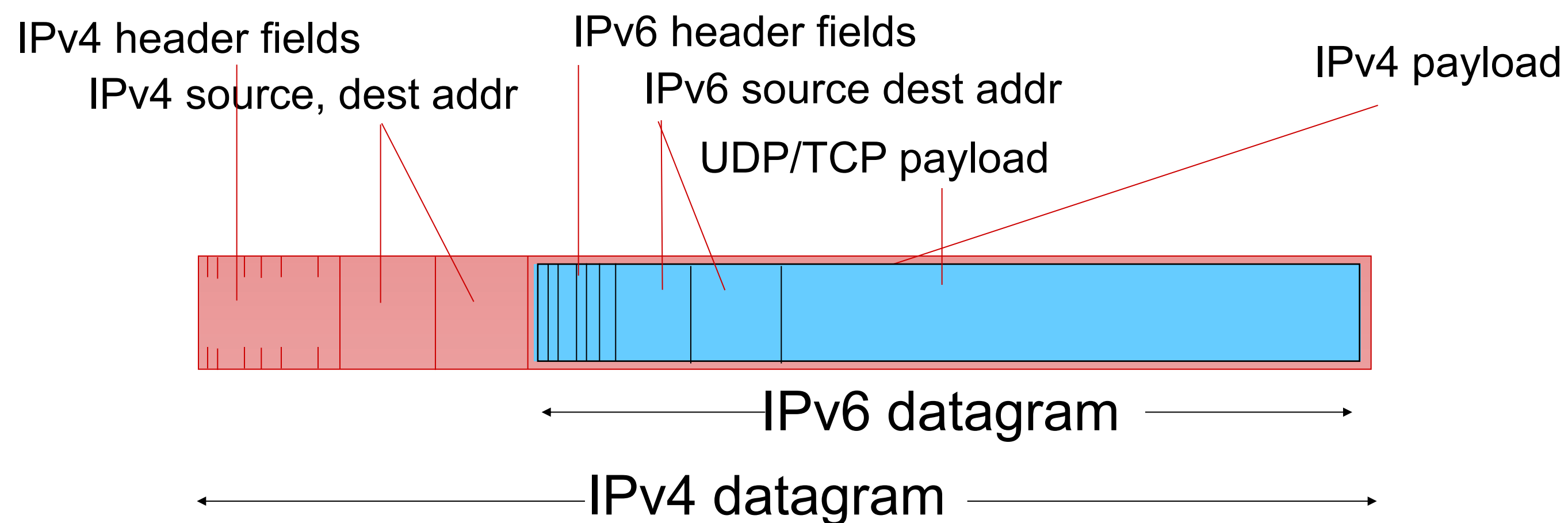
Flow label: identify
datagrams in same
"flow." (concept of
"flow" not well defined).

40 byte header

- ▶ What's missing (compared with IPv4):
 - ▶ No checksum (to speed processing at routers)
 - ▶ No fragmentation/reassembly
 - ▶ No options (available as upper-layer, next-header protocol at router)

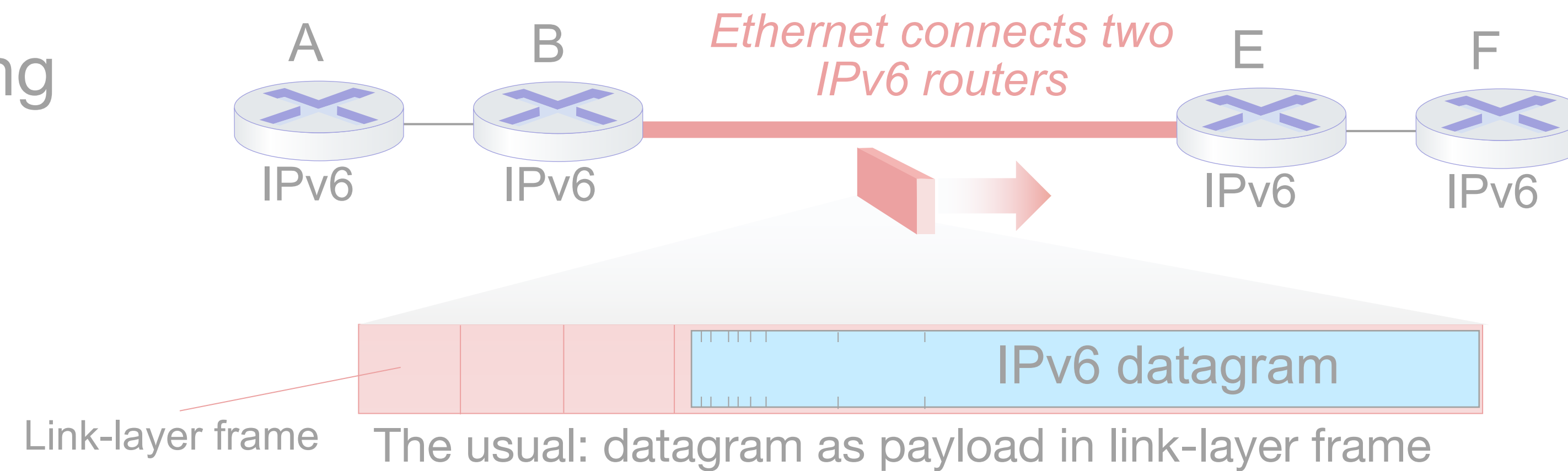
Transition from IPv4 to IPv6

- ▶ Not all routers can be upgraded simultaneously
 - ▶ No “flag days”
 - ▶ How will network operate with mixed IPv4 and IPv6 routers?
- ▶ Tunnelling: IPv6 datagram carried as payload in IPv4 datagram among IPv4 routers (“packet within a packet”)
- ▶ Tunnelling used extensively in other contexts (4G/5G)

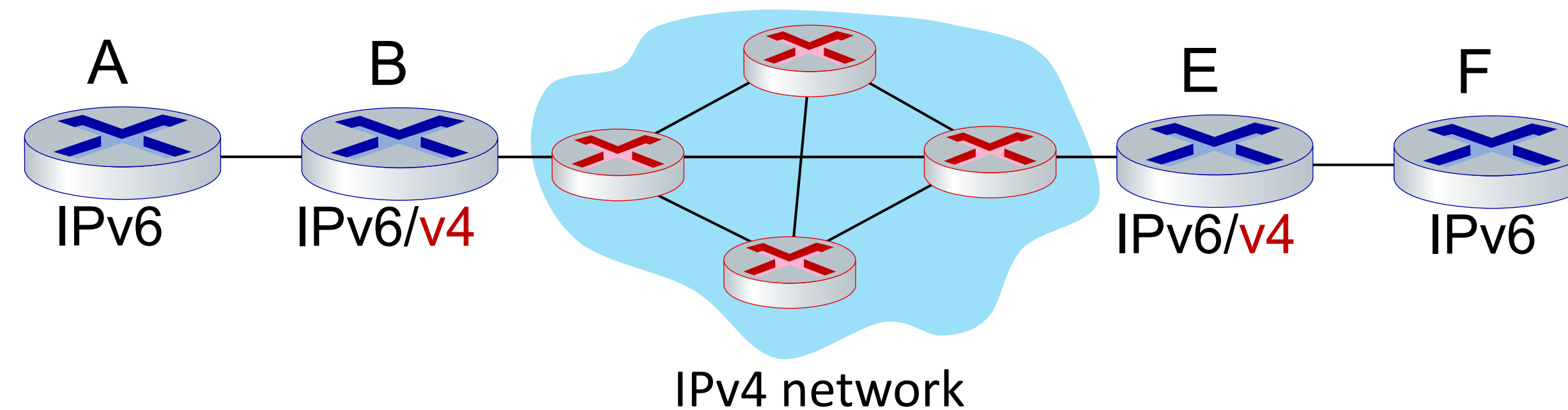


Tunnelling and encapsulation

Ethernet connecting two IPv6 routers

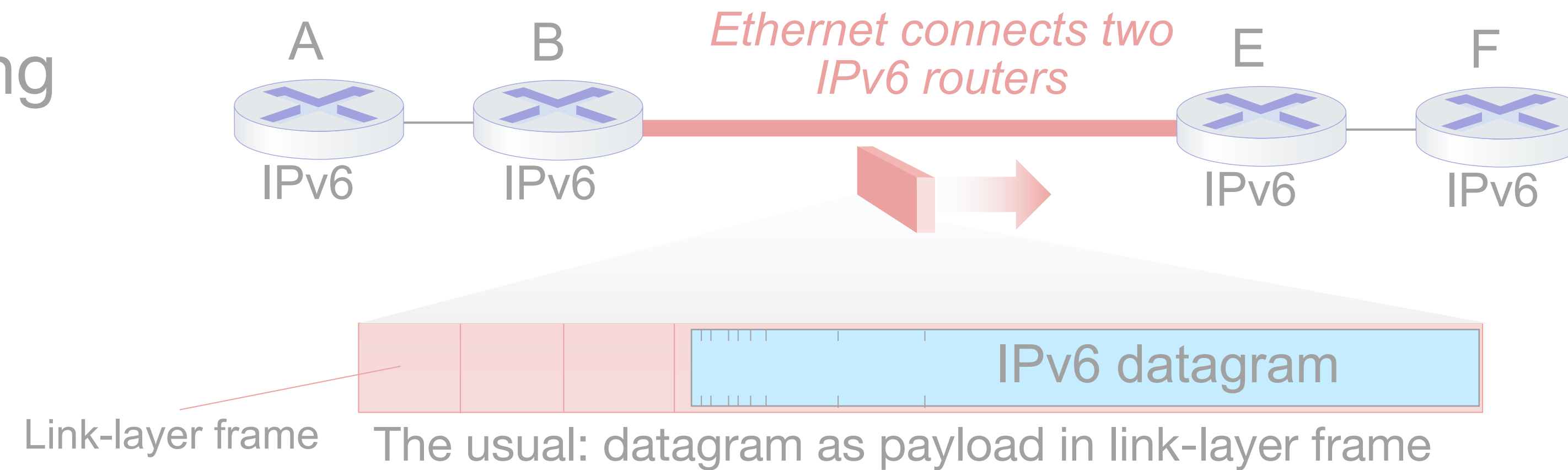


IPv4 network connecting two IPv6 routers

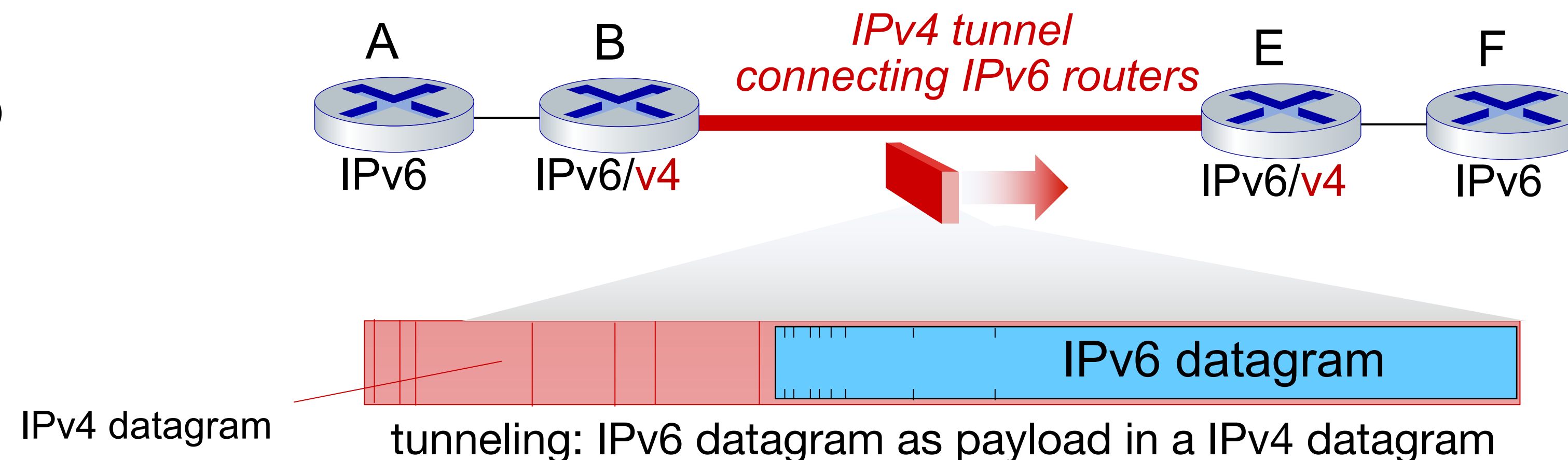


Tunnelling and encapsulation

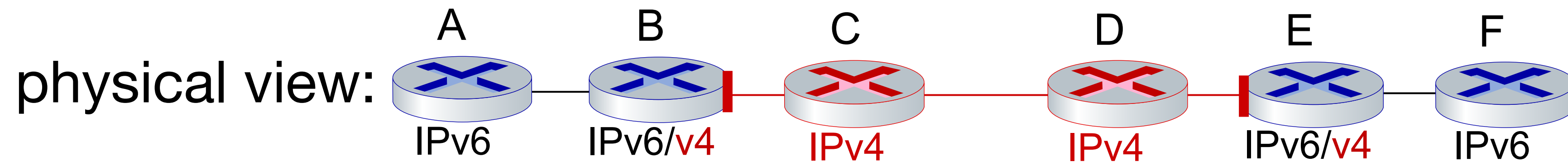
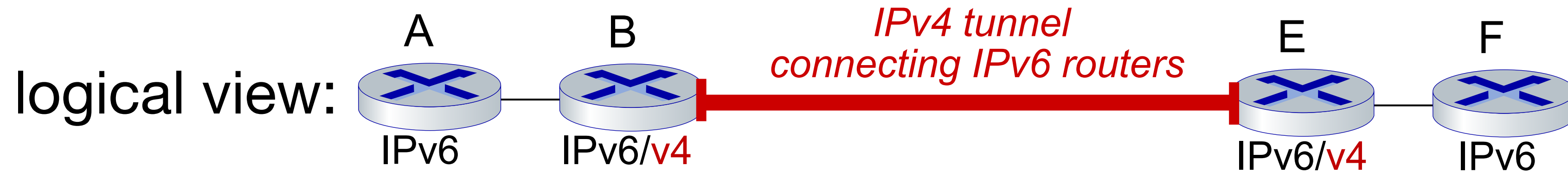
Ethernet connecting two IPv6 routers



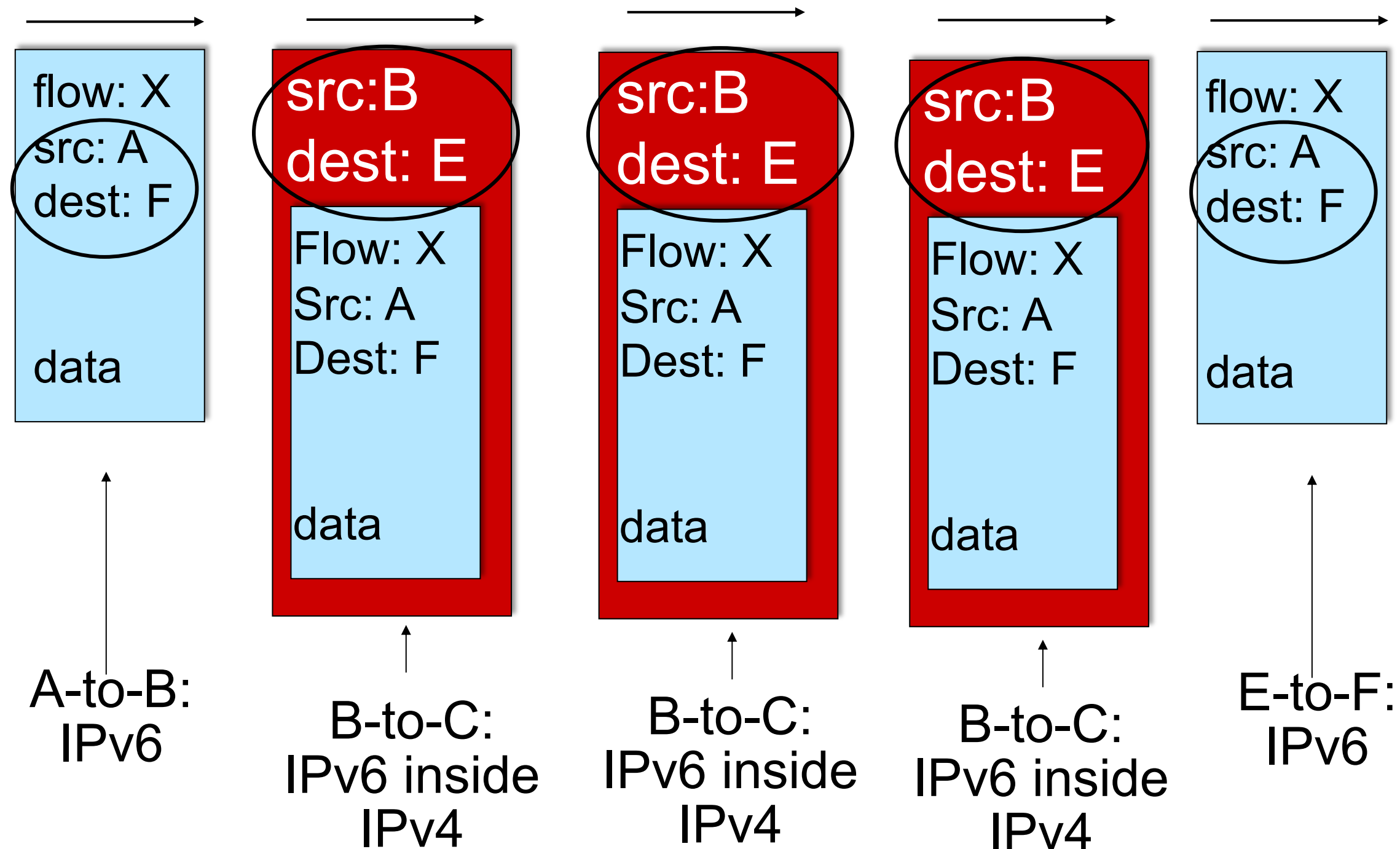
IPv4 network connecting two IPv6 routers



Tunnelling



Note source and destination addresses!



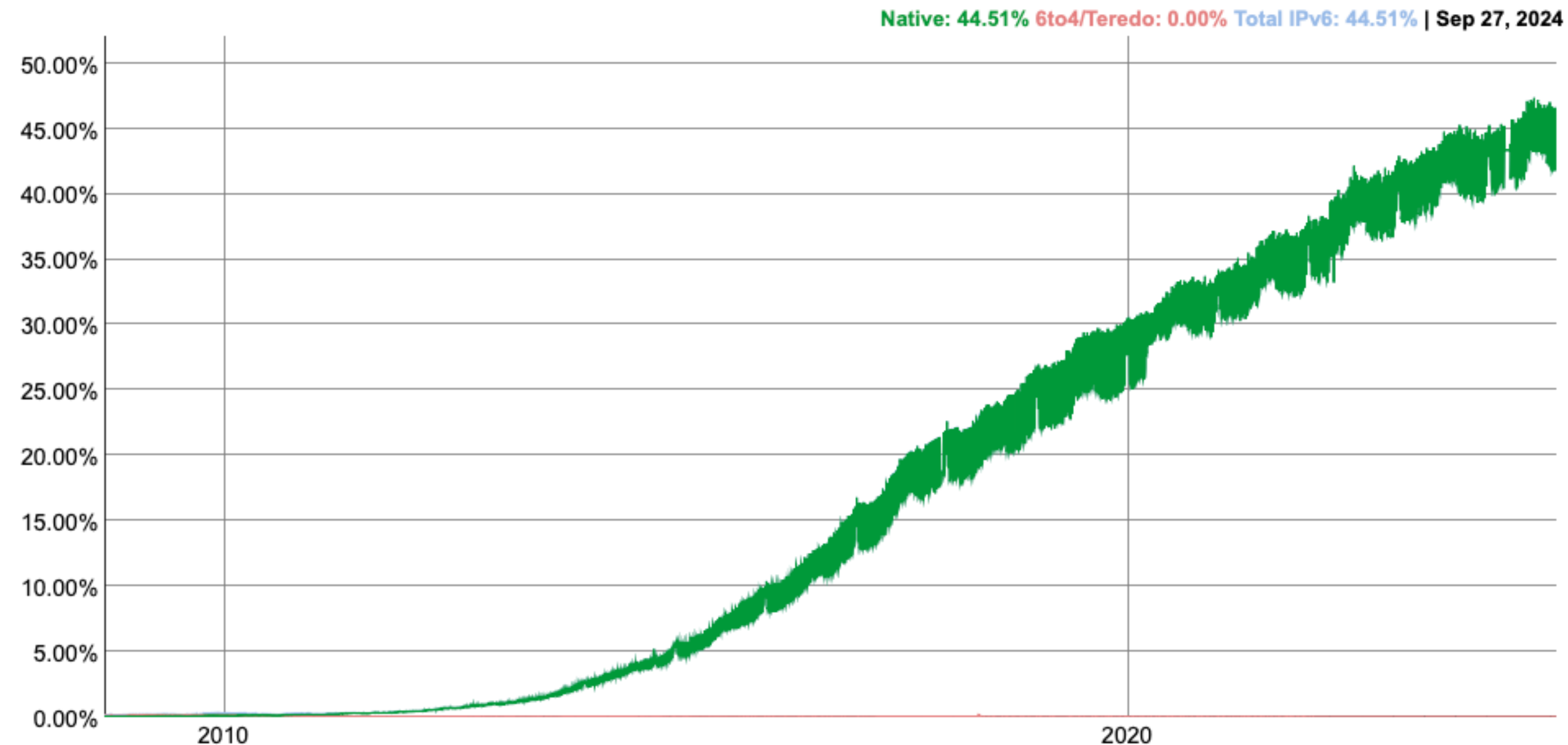
Note — This is also done the other way around, if client applications can only handle IPv4 and the network already uses IPv6

IPv6 – Adoption

► Google: $\geq 44\%$ of clients access services via IPv6 (2024)

IPv6 Adoption

We are continuously measuring the availability of IPv6 connectivity among Google users. The graph shows the percentage of users that access Google over IPv6.



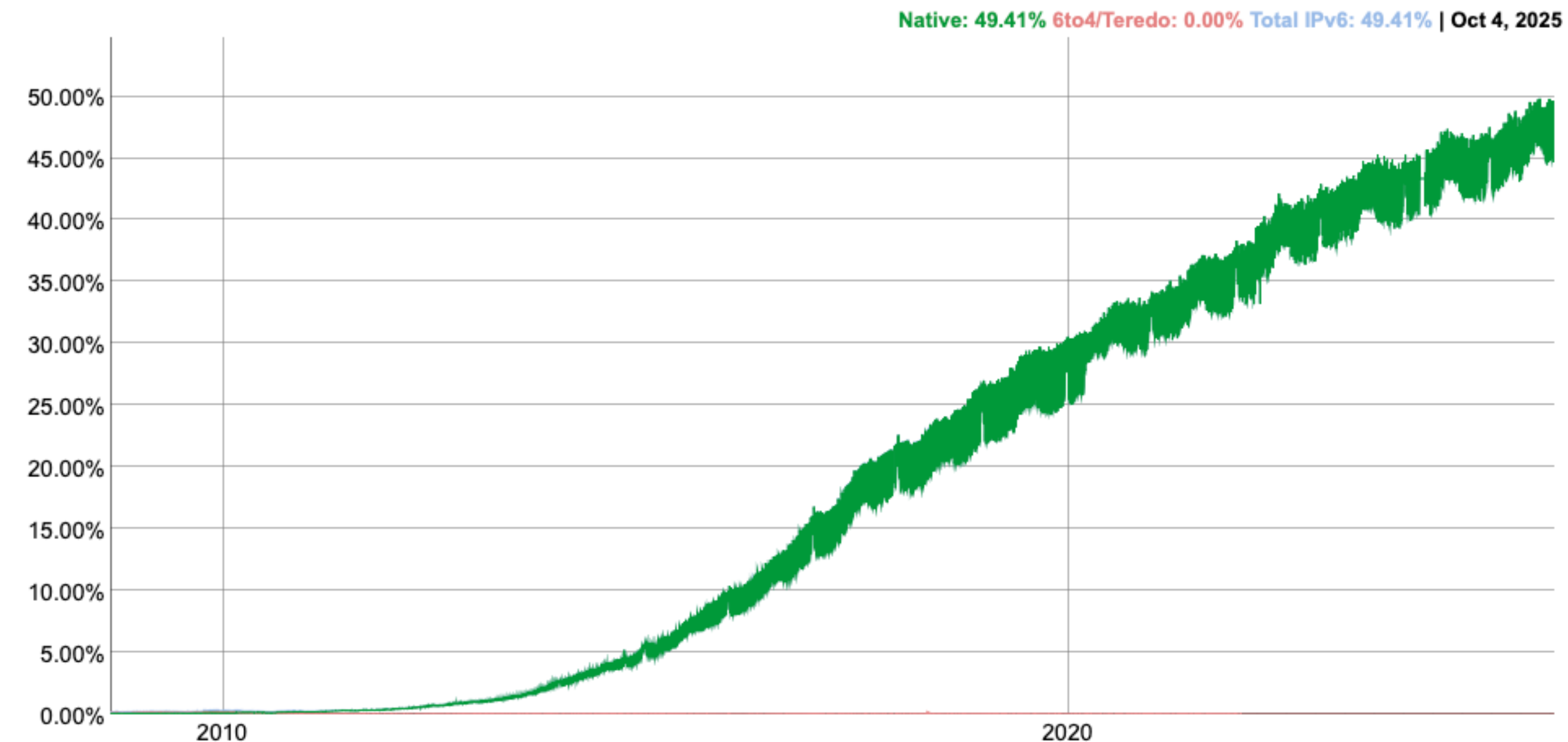
<https://www.google.com/intl/en/ipv6/statistics.html>

IPv6 – Adoption

► Google: $\geq 49\%$ of clients access services via IPv6 (2025)

IPv6 Adoption

We are continuously measuring the availability of IPv6 connectivity among Google users. The graph shows the percentage of users that access Google over IPv6.



~5% per year
currently

<https://www.google.com/intl/en/ipv6/statistics.html>

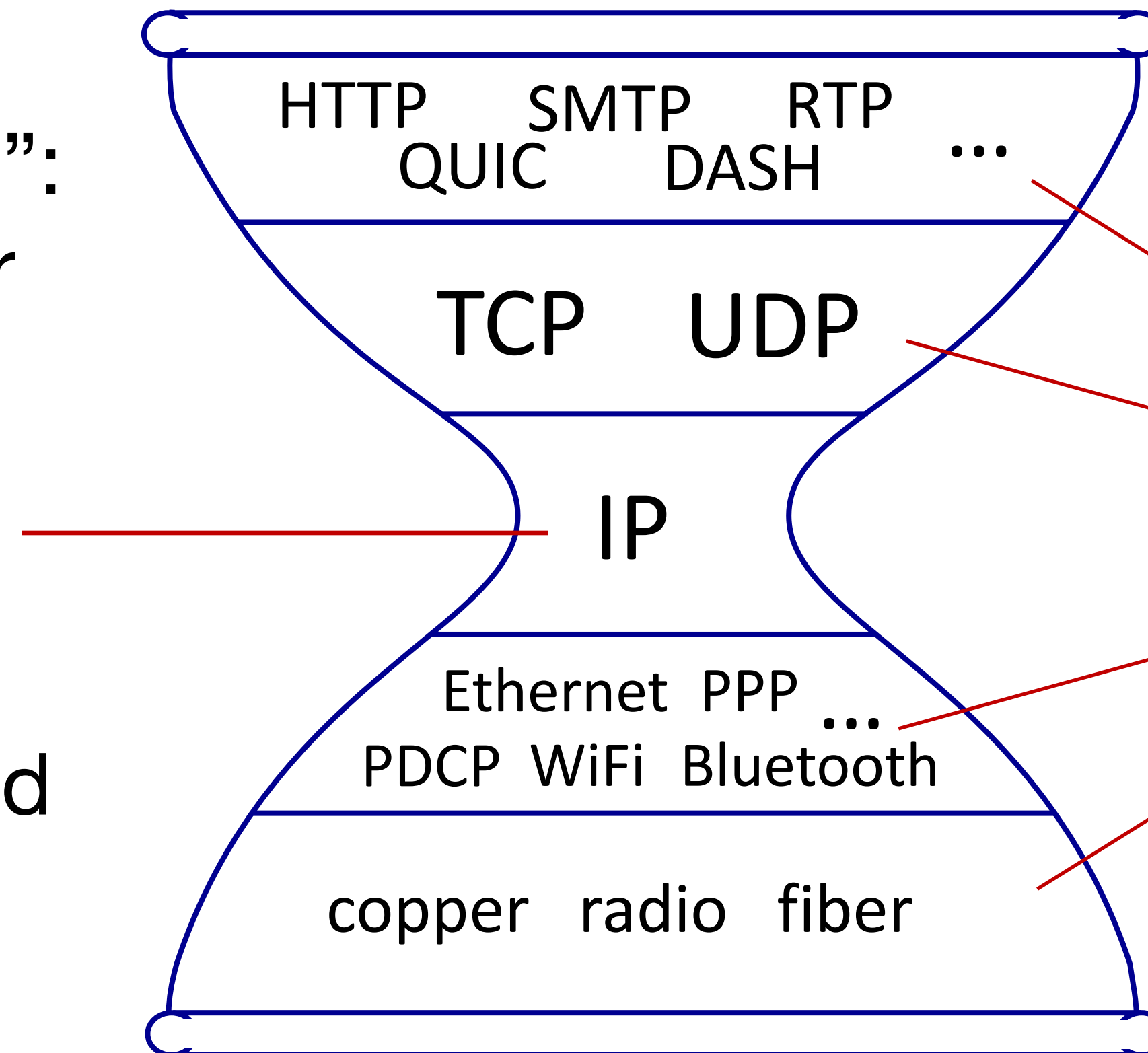
IPv6 – Adoption

- ▶ Google: $\gtrsim$ 49% of clients access services via IPv6 (2025)
- ▶ Long (long!) time for deployment, use
 - ▶ 25 years and counting!
 - ▶ Think of application-level changes in last 25 years: WWW, social media, streaming media, gaming, telepresence, ...
 - ▶ Why?

The IP hourglass

Internet's "thin waist":

- *One* network layer protocol: IP
- *Must* be implemented by every (billions) of Internet-connected devices



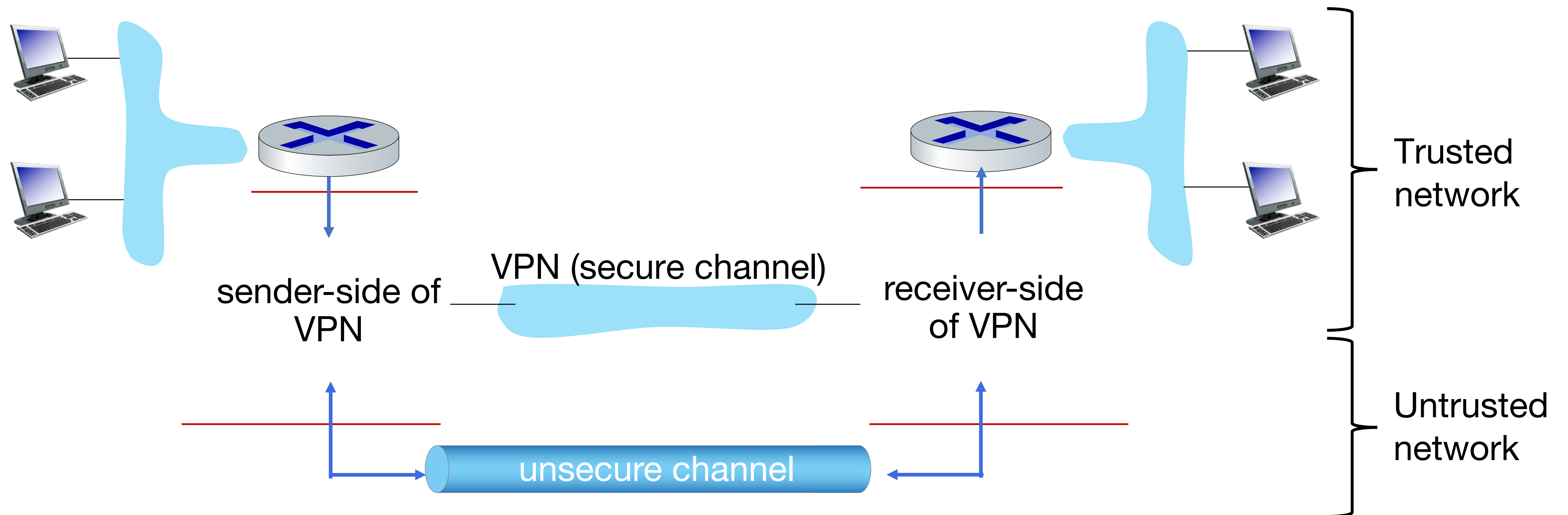
Many protocols in physical, link, transport, and application layers

Learning Goal 5A455F

What is IPsec?

What is a VPN?

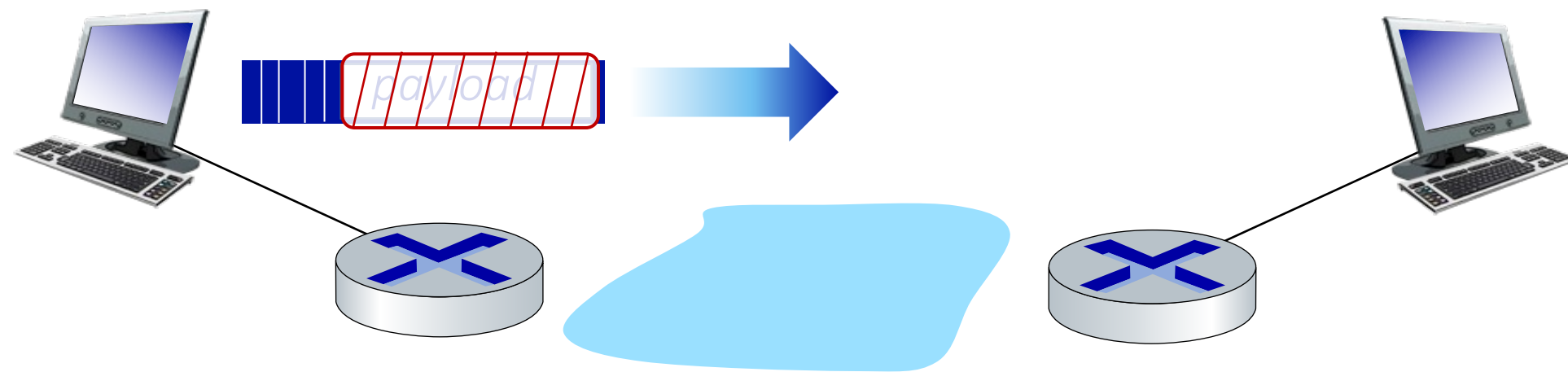
- ▶ Provide an abstraction to add security to the network layer
- ▶ Allows connecting hosts through an unsecure network, e.g., the Internet
- ▶ Different VPN protocols exist



IPsec

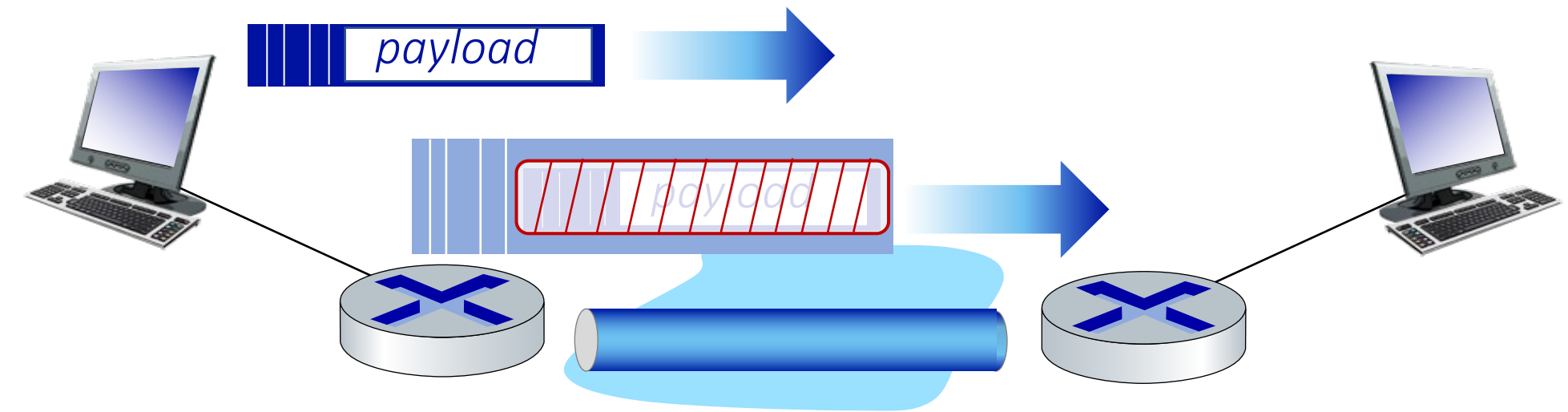
- ▶ Provides datagram-level encryption, authentication, integrity
 - ▶ For both user traffic and control traffic (e.g., BGP, DNS messages)

- ▶ Two “modes”



Transport Mode

- ▶ **Only** datagram **payload** is encrypted, authenticated



Tunnel Mode

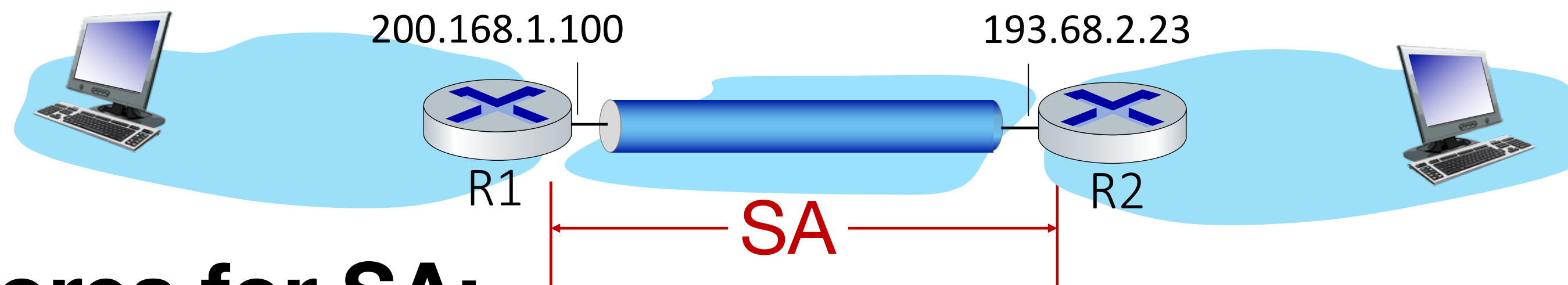
- ▶ **Entire datagram** is encrypted, authenticated
- ▶ Encrypted datagram encapsulated in new datagram with new IP header, tunneled to destination

Two IPsec Protocols

- ▶ **Authentication Header (AH) protocol [RFC 4302]**
 - ▶ Provides source authentication & data integrity but not confidentiality
- ▶ **Encapsulation Security Protocol (ESP) [RFC 4303]**
 - ▶ Provides source authentication, data integrity, and confidentiality
 - ▶ more widely used than AH

Security Associations (SAs)

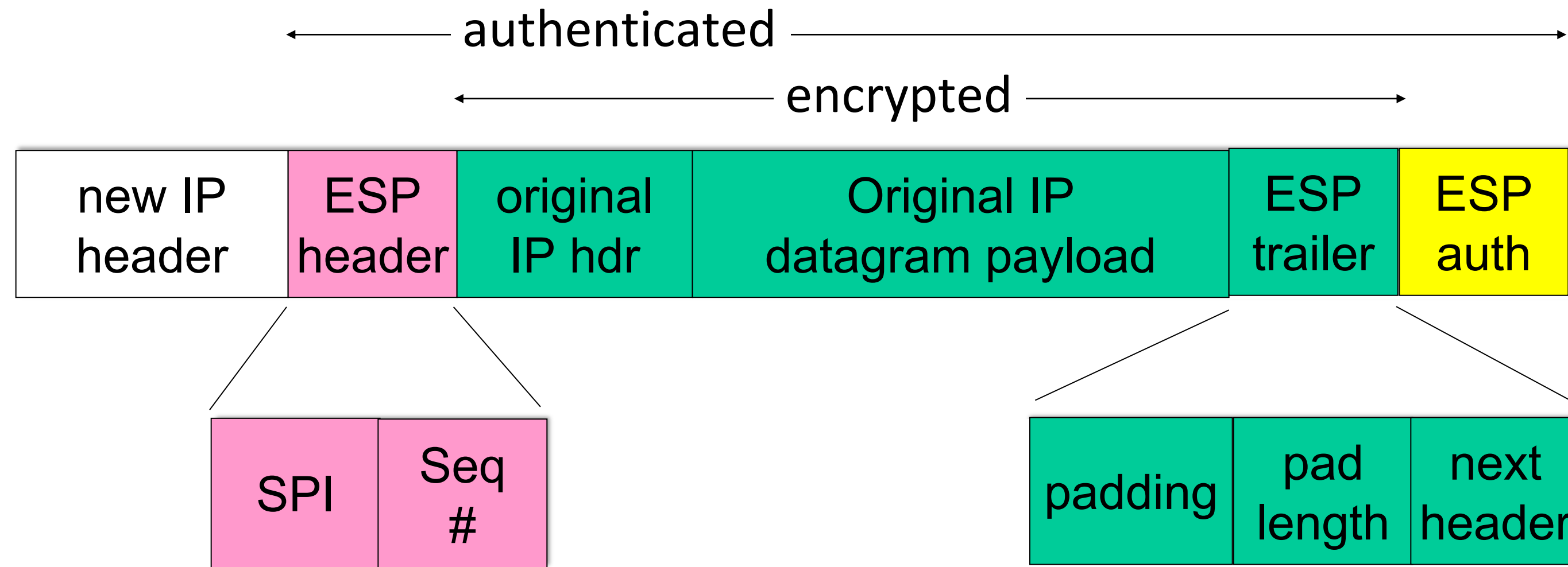
- ▶ Before sending data, security association (SA) established from sending to receiving entity (directional)
- ▶ Sending, receiving entities maintain state information about SA
 - ▶ Recall: TCP endpoints also maintain state info
 - ▶ IP is connectionless; IPsec is connection-oriented!



R1 stores for SA:

- ▶ 32-bit identifier: *Security Parameter Index (SPI)*
- ▶ origin SA interface (200.168.1.100)
- ▶ destination SA interface (193.68.2.23)
- ▶ type of encryption used
- ▶ encryption key
- ▶ type of integrity check used
- ▶ authentication key

IPsec Datagram



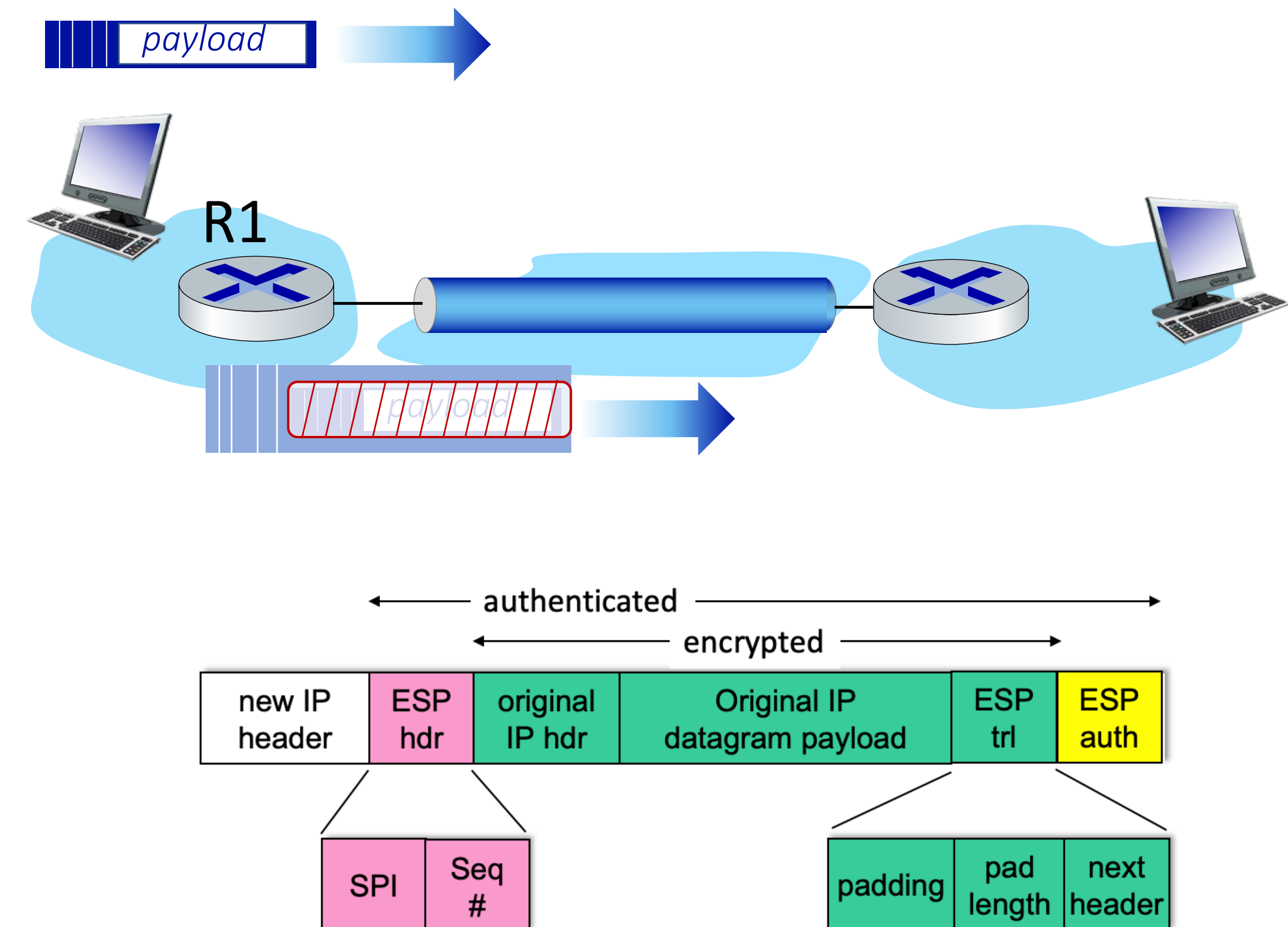
tunnel mode
ESP

- ▶ ESP trailer: padding for block ciphers
- ▶ ESP header
 - ▶ SPI, so receiving entity knows what to do
 - ▶ Sequence number, to thwart replay attacks
- ▶ MAC in ESP auth field created with shared secret key

ESP Tunnel Mode - Procedure

At R1

- ▶ Appends ESP trailer to original datagram (which includes original header fields!)
- ▶ Encrypts result using algorithm & key specified by SA
- ▶ Appends ESP header to front of this encrypted quantity
- ▶ Creates authentication MAC using algorithm and key specified in SA
- ▶ Appends MAC forming payload
- ▶ Creates new IP header, new IP header fields, addresses to tunnel endpoint



IPsec Sequence Numbers

- ▶ For new SA, sender initializes seq. # to 0
- ▶ Each time datagram is sent on SA:
 - ▶ Sender increments seq # counter
 - ▶ Places value in seq # field
- ▶ Goal
 - ▶ Prevent attacker from sniffing and replaying a packet
 - ▶ Receipt of duplicate, authenticated IP packets may disrupt service
- ▶ Method
 - ▶ Destination checks for duplicates
 - ▶ Doesn't keep track of all received packets; instead uses a window

IPsec Security Databases

Security Assoc. Database (SAD)

- ▶ Endpoint holds SA state in security association database (SAD)
- ▶ When sending IPsec datagram, R1 accesses SAD to determine how to process datagram
- ▶ When IPsec datagram arrives to R2, R2 examines SPI in IPsec datagram, indexes SAD with SPI, processing datagram accordingly

SAD defines how to do it

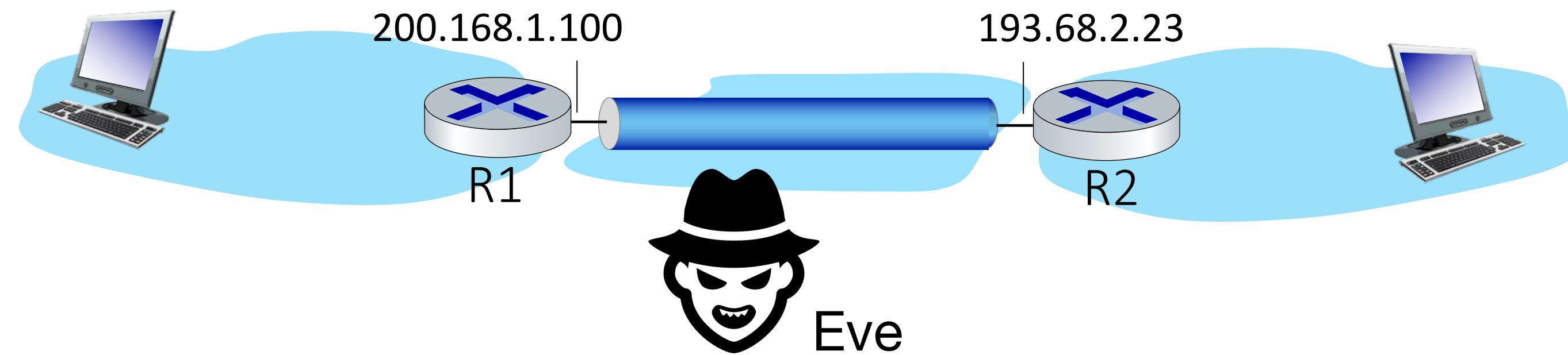
Security Policy Database (SPD)

- ▶ **Policy:** for given datagram, sender needs to know if it should use IPsec
- ▶ Policy stored in security policy database (SPD)
- ▶ Needs to know which SA to use
- ▶ May use: source and destination IP address; protocol number

SPD defines what to do

IPsec Services – Summary

- ▶ Eve sits somewhere between R1, R2
 - ▶ They don't know the keys



- ▶ Will Eve be able to
 - ▶ See original contents of datagram?
 - ▶ See source, destination IP address, transport protocol, application port?
 - ▶ Flip bits without detection?
 - ▶ Spoofing as R1 using R1's IP address?
 - ▶ Replay a datagram?

Internet Key Exchange (IKE)

- ▶ Previous examples: manual establishment of IPsec SAs in IPsec endpoints
- ▶ Example SA
 - ▶ SPI: 12345
 - ▶ Source IP: 200.168.1.100
 - ▶ Dest IP: 193.68.2.23
 - ▶ Protocol: ESP
 - ▶ Encryption algorithm: 3DES-cbc
 - ▶ HMAC algorithm: MD5
 - ▶ Encryption key: 0x7aeaca...
 - ▶ HMAC key: 0xc0291f...
- ▶ Manual keying is impractical for VPN with 100s of endpoints
- ▶ Instead use IPsec IKE (Internet Key Exchange)

IKE Modes – PSK and PKI

- ▶ Authentication (prove who you are) with either
 - ▶ Pre-shared secret (PSK) or
 - ▶ With PKI (public/private keys and certificates)
- ▶ PSK: both sides start with secret
 - ▶ Run IKE to authenticate each other and to generate IPsec SAs (one in each direction), including encryption, authentication keys
- ▶ PKI: both sides start with public/private key pair, certificate
 - ▶ run IKE to authenticate each other, obtain IPsec SAs (one in each direction).
 - ▶ similar with handshake in SSL

IKE Phases

- ▶ IKE has two phases
 - ▶ **Phase 1:** Establish bi-directional IKE SA (aka “ISAKMP security association”)
 - ▶ Note: IKE SA different from IPsec SA
 - ▶ Has two modes: aggressive mode and main mode
 - ▶ Aggressive mode uses fewer messages
 - ▶ Main mode provides identity protection and is more flexible
 - ▶ **Phase 2:** Internet Security Association and Key Management Protocol (ISAKMP) is used to securely negotiate IPsec pair of SAs
- Once again the (in this course elusive) Diffie-Hellman key exchange protocol is used here.

This will not be in the exam

Adapted from slides by J.F Kurose and K.W. Ross

IPsec Summary

- ▶ IKE message exchange for algorithms, secret keys, SPI numbers
- ▶ Either AH or ESP protocol (or both)
 - ▶ AH provides integrity, source authentication
 - ▶ ESP protocol (with AH) additionally provides encryption
- ▶ IPsec peers can be two end systems, two routers/firewalls, or a router/firewall and an end system

Learning Goals

- ▶ Get an understanding of routing on the network layer ✓
 - ▶ Input port functions
 - ▶ Longest prefix matching
 - ▶ Buffer management
 - ▶ Scheduling policies
- ▶ Understand the Internet Protocol and network addressing ✓
 - ▶ IPv4 basics
 - ▶ Addressing & subnets
 - ▶ DHCP
 - ▶ NAT
 - ▶ IPv6

Learning Goals

- ▶ Understand how IPsec can be used to establish a VPN ✓
 - ▶ IPsec basics
 - ▶ Security Associations
 - ▶ Security Policy & Security Association Databases
 - ▶ Internet Key Exchange